# Group flights in free flight

Delayed contacts, configurations and episodes:
a method to label every instant of every flight

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**Abstract**

We describe a method that decides, for every flight and every instant, whether a paraglider pilot flies in a group or alone and, in the first case, with which other pilots. Two pilots are in contact when one of them is where the other was, or will be, a few minutes earlier or later. This is the proximity with a local delay of Li, Wu and Crofoot [15], adapted to thermal flight in three ways: a vertical tolerance separates pilots who circle in the same thermal at distant heights; headings are compared only in glide; the maximum delay is bounded by the visual range and measured on the data. At each instant the groups are the connected components of the contact graph. As in the grouping structure of Buchin et al. [6], a configuration is a fixed set of flights over a maximal interval, and an episode is a chain of configurations that share members. Contacts are found in two stages, a coarse filter that provably loses none and an exact scan; their minimum duration is calibrated on a null model that shifts tracks in time. On the FFVL archive most short contacts are chance: the maximum delay is measured at 2 min, and a contact must last 38 min for at most one in ten to be chance. The output is a few tables that label every instant of every flight, so that group flight can be compared with solo flight.

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## The method at a glance

Figure 1 follows the method from two tracks to the units of the statistical analysis, and Table 1 says what each object is in plain words, with the section that defines it. Figures 7, 12 and 18, marked ▶ , are animated: in Adobe Acrobat Reader, or another viewer that runs the JavaScript of PDF files, the buttons below them play the animation; other viewers show the figure as printed. The same animations are GIF files in the folder `animations/`.

## 1 Motivation and requirements

The statistical properties of free-flight trajectories (time spent in thermals, length of the glides, mean squared displacement) are usually estimated by treating flights as independent samples. The assumption fails when several pilots fly together: they share thermals and routes, and their trajectories are correlated. Group flight may also change the behaviour itself. In soaring

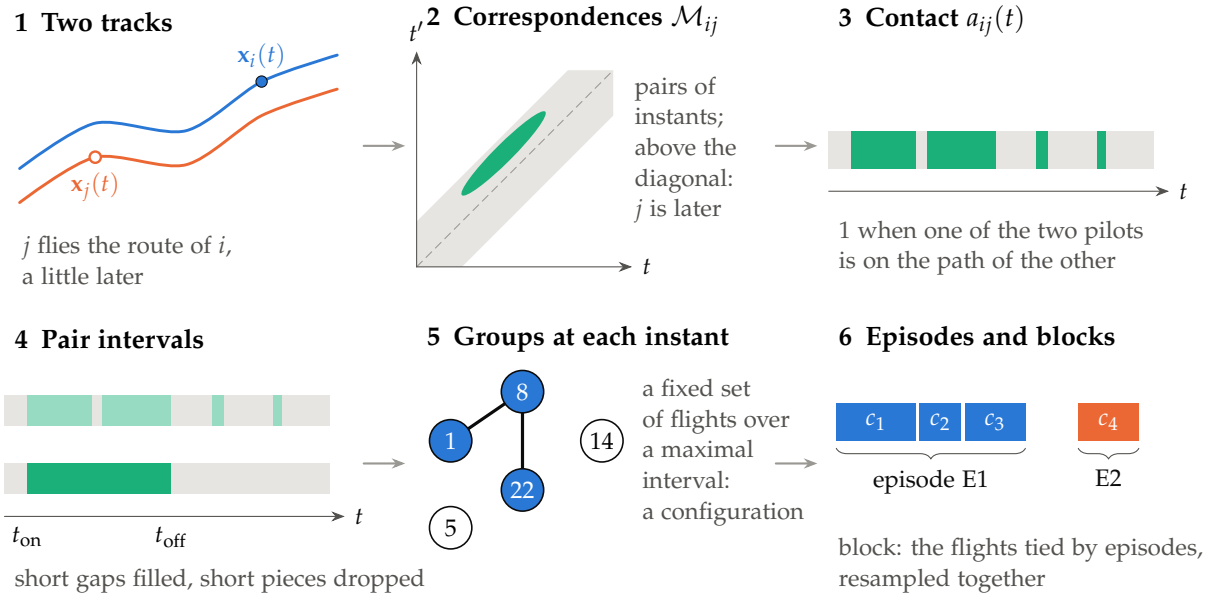



**Figure 1:** The method in six steps. Panels 1–3: the pair contact (Section 4). Panels 4–6: from pairs to groups (Section 5). Flights 1, 8, 22, 14 and 5 are the example day of Figure 18.

| Object | In words | Symbol | Section |
|---|---|---|---|
| Departure | the first fix farther than $R_{\text{dep}}$ from the takeoff | $k_i^{\text{dep}}$ | 4.1 |
| Correspondence | two instants of the same clock, $t$ for $i$ and $t'$ for $j$, at most $\delta$ apart, at which the two pilots pass the same place within the tolerances | $(t, t') \in \mathcal{M}_{ij}$ | 4.2 |
| Contact | at instant $t$, one of the two pilots is on the path of the other | $a_{ij}(t) = 1$ | 4.2 |
| Pair interval | a stretch of contact, with short gaps filled and short pieces dropped | $[t_{\text{on}}, t_{\text{off}})$ | 5.1 |
| Group at instant $t$ | the pilots joined to each other by chains of contacts at $t$ | component of $G(t)$ | 5.3 |
| Configuration | a fixed set of flights that stays one group over a maximal interval | $c$ | 5.3 |
| Membership | the label of flight $i$ at instant $t$: local phase, solo, or its configuration | $\gamma_i(t)$ | 5.3 |
| Episode | a chain of configurations, each starting when the previous one ends and sharing a member with it | | 5.4 |
| Dependency block | the flights tied together by episodes; the bootstrap resamples them together | | 5.5 |

**Table 1:** A minimal dictionary of the objects of the method.

birds, social information on the position of thermals changes the gliding strategy [28], and in a migrating flock a few individuals find the thermals while the others follow them [8, 19]. Comparing group flights with solo flights therefore needs a rule that says, for every flight and every instant, whether the pilot is in a group and with whom.

The intuitive notion of *flying together* that the rule must make operational is the following (Figure 2).

(R1) The pilots take off from the same area, not necessarily from the same takeoff.

(R2) Departures may be staggered, but not to the point of losing sight of each other.

(R3) The pilots follow roughly the same route: nearby places and similar motion, not necessarily the same points.

(R4) The composition of the group changes in time: some pilots join, others change route.

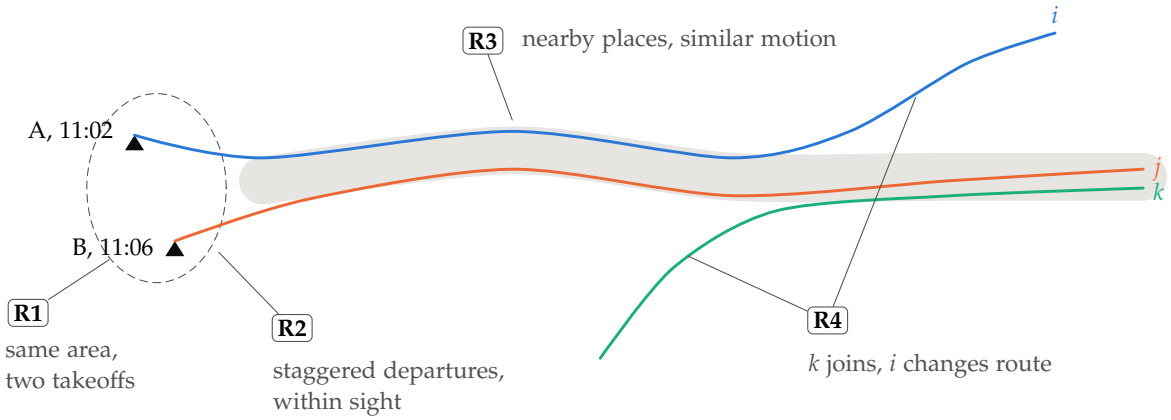



**Figure 2:** The four requirements on a schematic map. Pilots $i$ and $j$ take off from two places of the same area four minutes apart and follow the same corridor (grey) without flying over the same points; $k$ joins later and $i$ leaves the corridor.

Three immediate approaches do not meet these requirements (Figure 3).

- **Geometric similarity of the tracks** (Hausdorff, Fréchet, unconstrained DTW; survey in [23]). Cross-country routes are channelled by the terrain and thermals are often anchored to the ground: two independent pilots who took off from the same site produce similar tracks even hours apart. Unconstrained DTW, moreover, warps the time axis without limit and erases exactly the information that (R2) needs.
- **Strict synchrony** (flock [10], convoy [11], swarm [14]). These patterns require proximity at the same instant; the swarm admits non-consecutive instants but no delay between members. Two pilots on the same route two minutes apart are never close at the same instant, yet they fly together in the sense of (R2)–(R3).
- **Window on the departures.** A rule such as "departures within $T$ minutes" produces groups that depend on the reference pilot: if 1 and 8 depart within the window, and so do 8 and 14, then 1 and 14 need not (Section 5.2).

## 2 Relation to the literature

This section says where each ingredient of the method comes from and what, as far as a targeted but not exhaustive search shows, has not been done yet.

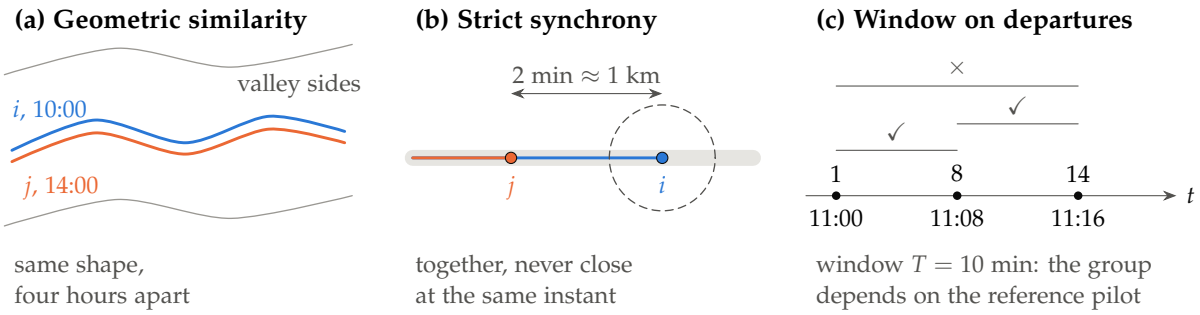



**Figure 3:** Why three immediate rules fail. (a) Two independent pilots on the same valley produce similar tracks hours apart. (b) Two pilots on the same route two minutes apart never enter each other's proximity disc. (c) With a window of 10 minutes, 1–8 and 8–14 pass and 1–14 fails.

## 2.1 Trajectory similarity with a temporal tolerance

The LCSS of Vlachos, Kollios and Gunopulos [27] matches two points when they are closer than $\varepsilon$ in every coordinate and their *indices* in the two sequences differ by at most $\delta$. The similarity $S_1$ is the length of the common subsequence divided by the length of the shorter track, and the variant $S_2$ maximises $S_1$ over all spatial translations, to recognise parallel movements. The EDR of Chen, Özsu and Oria [7] and DTW with a Sakoe–Chiba band [21] bound the shift in the same way. These measures return a single number per pair of tracks: they say how similar two tracks are, and not *when* two pilots are together. A bound on indices counted from the first point of each track also mixes the shift between the takeoffs with the delay in flight, and $S_2$ matches precisely the parallel routes that we want to exclude.

The measures of joint movement that do say when are built on simultaneous fixes. The proximity index Prox is the share of simultaneous fixes closer than a threshold [12, §3.1]: with $\delta = 0$ and without the conditions on height and heading, it is the share of the flight of $i$ in contact with $j$ in the sense of Section 4.2.

## 2.2 Following with a delay

The closest precedent of our pair contact is the following of Li, Wu and Crofoot [15], defined on tracks sampled at the same instants (Figure 4). At instant $i$ the local lag $\Delta(i)$ is the shift $t$, with $|t| \leq l_{\max}$, that minimises the distance between the position $r_i$ of $R$ and the position $s_{i-t}$ of $S$. The instant is *valid* if this minimum distance is below $d_{\max}$, and $R$ follows $S$ at a valid instant if $\Delta(i) > 0$ (§IV-A). The lag is therefore local and free to vary, and its sign separates the follower from the leader. Two individuals that merely move together lead and follow equally often, so the following intervals are the stretches with more following instants than the half expected without following: the maximal segments of $f(i) - 1/2$, with $f(i) = 1$ when $R$ follows and 0 otherwise, found in linear time [20] (§IV-B) and checked with the roles reversed (§IV-C). The threshold $d_{\max}$ only discards the instants without interaction, and does not move the minimiser.

Our contact keeps the window of lags and the distance threshold, and leaves the sign out: the contact of $i$ with $j$ at instant $t$ is the validity of $t$, with tolerances on height and heading added (Section 4.2). We want the pairs that fly together, whether or not one of them leads; who follows whom comes back afterwards, through the median lag of each contact (Section 4.5).

Nagy et al. [18] measure the delay between two pigeons as the shift that maximises the correlation of their headings, and build leader–follower hierarchies from its sign. In a thermal the heading turns through $2\pi$ every 20–30 s, so this correlation, like the distance of Section 4.3, repeats with the period of the turn. Konzack et al. [13] display such local delays between simultaneously recorded tracks in a "delay space", comparing Fréchet, DTW and edit-distance

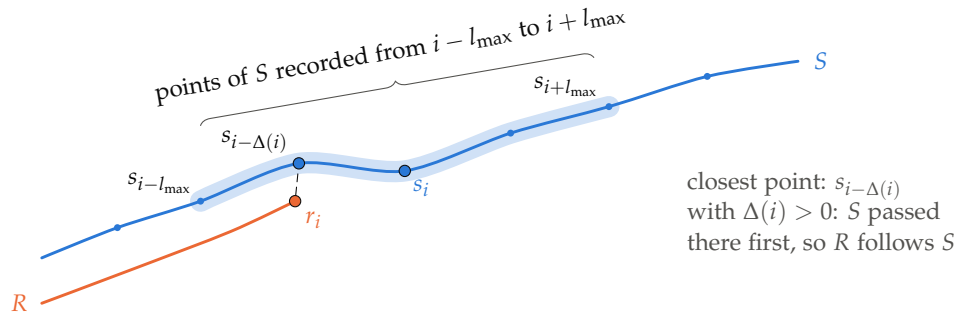



**Figure 4:** Following with a local lag [15]. At instant $i$, the track of $S$ is searched in the time window $[i - l_{\max}, i + l_{\max}]$ (shaded) for the point closest to the position $r_i$ of $R$. The best lag $\Delta(i)$ is positive: $R$ is now where $S$ was $\Delta(i)$ earlier.

matchings. FLICA and its package mFLICA [2, 3, 4] infer a following network in sliding time windows, with DTW restricted to a Sakoe–Chiba band, and call the faction of a leader the individuals that reach it through a directed path: groups built from delayed following, but directed, with leaders, and on windows. The following of Andersson et al. [5] is instantaneous: an individual follows another that is in a sector ahead of it with a similar heading.

### 2.3 Group structure in time

Buchin et al. [6] call two entities directly connected at time $t$ if their $\varepsilon$-discs overlap, and $\varepsilon$-connected if a chain of direct connections joins them at that instant; the maximal $\varepsilon$-connected sets, the components, partition the entities at every instant. The Reeb graph records how the components evolve: its vertices are the events (start, end, merge, split), and each edge is a fixed set of entities that forms one component over an interval of time (Figure 5). On it they define the maximal groups, sets of at least $m$ entities that stay inside one component for at least a duration $\delta$; maximal groups may overlap, and an entity can belong to several at once. Two more ideas of theirs reappear below: a robust structure ignores interruptions shorter than $\alpha$, by calling two entities connected at $t$ when they were directly connected at some instant within $\alpha/2$ of $t$ (§4); and the grouping structure can split the trajectories into independent data sets (§1). The proximity is always simultaneous.

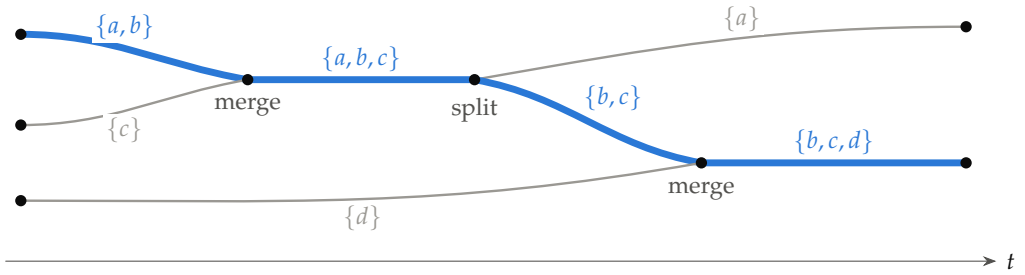



**Figure 5:** The grouping structure of Buchin et al. [6] as a Reeb graph. Each edge is a fixed set of entities that forms one connected component over an interval; vertices are starts, ends, merges and splits. The thick edges have at least two entities: in this note they become the configurations (Section 5.3), built on delayed contacts.

### 2.4 Interaction indices and null models

Joo et al. [12] sort the metrics of joint movement between pairs by what they capture, proximity or coordination. Long et al. [16] compare eight of them on simulated tracks and on deer: the indices that rest on a statistical test report interactions that are not there, more often the finer the sampling, and the indices that return one number per pair miss the interactions that come and go; the wildlifeDI package [17] extends the analysis to contacts in high-frequency data. Spiegel et al. [22] build a null model by shuffling whole days of each individual's path among its days: each individual keeps its use of space, and the synchrony between individuals is lost.

### 2.5 Collective soaring

Ákos, Nagy and Vicsek [1] compare the soaring of birds with that of paraglider and hang-glider pilots on GPS tracks; falcons, storks and paragliders circle with radii of 20–26 m. In migrating white storks a few leaders find and explore the thermals and the others follow them, leaving the thermals earlier and lower [8]; inside a thermal, flock mates synchronise their circling [19]. Vultures informed by other birds of the next updraft glide towards it faster [28]. We found no work that identifies group flights systematically in large free-flight datasets.

### 2.6 What we take and what is new

The method is not a new general algorithm for grouping trajectories. The pair contact is the proximity with a local lag of Li et al. [15], the group structure is that of Buchin et al. [6], and groups built from delayed relations exist in mFLICA [4], with other definitions (Table 2). What we did not find in the literature is:

(i) the Reeb structure built on contacts with a local delay instead of simultaneous proximity;

(ii) the adaptations to thermal flight: a vertical tolerance that separates pilots circling in the same thermal at distant heights, and headings compared only in glide (Section 4.3);

(iii) the application to a free-flight dataset of this size, with thresholds calibrated on a null model.

The contribution is therefore a physically motivated adaptation and a scalable implementation, more than a new method in itself, and the thesis should present it in these terms.

**Essential reading.** The two readings to do first are Li et al. [15], Sections II and IV (following, local lag, score), and Buchin et al. [6], Sections 1, 2.1 and 4 of the arXiv version (groups, Reeb graph, robustness). Long et al. [16] and Spiegel et al. [22] serve the null model; Vlachos et al. [27] and Joo et al. [12] place the measures of similarity and proximity.

## 3 Data and notation

The flights are those of the xc-dynamics pre-processing. Each flight $i \in V$ is one or more segments of fixes, from its takeoff $k_i^0$ to its landing $k_i^{\text{fin}}$ (step 8 of the pre-processing, take-off and landing); between two segments the track has a gap. A fix has a time, a latitude $\varphi$, a longitude $\lambda$ and the GNSS altitude $z$, at the interval of its logger: one second for 70% of the paraglider flights, two to ten seconds for most of the others. Time $k$ is in UTC seconds from the start of the day, on the same clock for all flights; IGC times are whole seconds.

| Component | Precedent | What we take | What we change |
|---|---|---|---|
| Pair contact with a delay | Li et al. [15]; tolerance in [27, 7, 21] | window of lags and distance threshold; local lag, free to vary | height tolerance (separates heights within a thermal); heading only in glide; local phase excluded; no sign in the contact, $a_{ij} = a_{ji}$ (the sign returns in $\ell_{\text{med}}$) |
| Contact intervals | robustness in [6]; maximal segments in [15] | tolerance to short interruptions | gaps shorter than $g$ filled; minimum duration calibrated on a null model |
| Groups in time | Buchin et al. [6] | components and their Reeb graph; configurations are its edges with at least two entities | built on delayed contacts; episodes are the connected parts of the Reeb graph without single entities; no maximal groups |
| Groups from following | mFLICA [4] | – | undirected groups without leaders; exact intervals instead of sliding windows |
| Null model | Spiegel et al. [22]; SAM [26] | time shift of the tracks; false discovery rate as chance count over observed count | shift within the day, drawn from the real delays between takeoffs; chance count per unit of exposure |
| Computation | filter–refine scheme | – | no-loss lemma for delayed contacts; KD-tree with a time coordinate |

**Table 2:** Where the ingredients of the method come from.

Positions follow the frame of the thesis (chapter Trajectory pre-processing, section Local coordinates and vertical datum): each fix is moved to the point of the WGS84 ellipsoid below it, $h = 0$, and converted to Earth-centred Cartesian coordinates, $\mathbf{x}_i(k) \in \mathbb{R}^3$. The thesis then rotates them into the east and north of the first fix of each flight; here they stay Earth-centred, so the coordinates of all flights are the same frame. The distance between two positions is the chord of the ellipsoid, shorter than the distance along it by about $d^3/24R^2$: about a millimetre at $d = 10$ km. The altitude is $z$ as recorded.

The phases of the run reported here are Vilpellet's runs, computed by the old repository (soaring, tag `legacy-final`) on the flights logged every second only, 70% of the paraglider flights. The phase labelling of xc-lab (note *Flight phases with a hidden Markov model*), meant for loggers of 5 s or less, was not used for this run. Every labelled fix of a flight has one of three states, $s_i(k) \in \{\mathsf{climb}, \mathsf{search}, \mathsf{transition}\}$: climbing in a thermal, searching for the next one, gliding between two thermals. In the text *glide* means the state transition. The fixes of all the other flights and of pieces too short to be labelled stay unclassified.

# 4 The pair contact

## 4.1 Local phase

Near the takeoff every pilot is close to every other, ridge soaring or waiting for the thermals, and a contact in this phase says nothing about (R3). The departure is the first fix farther than

| Symbol | Meaning | Start | Chosen |
|---|---|---|---|
| $\mathbf{x}_i(k) \in \mathbb{R}^3$ | position: the point of the ellipsoid below the fix, Earth-centred | | |
| $z_i(k)$ | GNSS altitude | | |
| $s_i(k)$ | state: climb, search, transition (glide), or unclassified | | |
| $\hat{\mathbf{v}}_i(k)$ | horizontal heading, used only in glide | | |
| $k_i^0$, $k_i^{\text{fin}}$, $k_i^{\text{dep}}$ | takeoff, landing, departure (Section 4.1) | | |
| $T_i^{\text{dep}}$ | the grid instants from departure to landing at which flight $i$ has a position | | |
| $R_{\text{dep}}$ | radius of the local phase around the takeoff | 3 km | 3 km |
| $\Delta$ | step of the common time grid $\Delta\mathbb{Z}$ | 5 s | 5 s |
| $h_{\max}$ | longest time between two fixes across which a grid position is interpolated | 10 s | 10 s |
| $\delta$ | maximum delay between two pilots, always a time | 5 min | 2 min |
| $\varepsilon_h$, $\varepsilon_z$ | horizontal and vertical tolerances | 500 m, 300 m | 500 m, 300 m |
| $\theta_{\max}$ | maximum angle between the headings in glide | 45° | 45° |
| $g$ | longest gap filled in a contact | 120 s | 120 s |
| $\tau^*$ | minimum duration of a contact | from the null model | 38.1 min |
| $v_{\text{xc}}$ | mean cross-country speed, measured on the data | $\approx 5$ m/s | |
| $v_{\max}$ | top speed between two fixes, set by the pre-processing (`speed.max_kmh`) | 100 km/h | 100 km/h |
| $\varepsilon_{\text{vis}}$ | visual range | 1.5 km | |
| $r_{\min}$, $R_s$ | smallest time shift and radius of the takeoffs in the null model | $> 2\delta$, 10 km | $2\delta + \Delta$, 10 km |
| $q$ | false discovery rate accepted for the contacts | 0.05 | 0.1 |

**Table 3:** Symbols and parameters, with their starting values and the values chosen on the data (`method_assessment.pdf`). Beware: in [27] $\delta$ is a tolerance on indices, in [12] a distance threshold, in [6] a minimum duration; here it is always a delay.

$R_{\text{dep}}$ from the takeoff:

$$k_i^{\text{dep}} = \min\{k : \ \|\mathbf{x}_i(k) - \mathbf{x}_i(k_i^0)\| > R_{\text{dep}}\}. \tag{1}$$

From then on the flight is evaluated on $T_i^{\text{dep}}$, the instants of the common grid, one every $\Delta$ seconds, from the departure to the landing at which the flight has a position (Section 6.1). The instants before the departure are not evaluated, and a flight that never leaves the disc is *local*. Both get the state L and do not count as solo flight (Figure 6).

## 4.2 Correspondences and contact

**Definition 4.1** (correspondence)**.** For two flights $i \neq j$, the set of correspondences is

$$\begin{aligned}\mathcal{M}_{ij} = \Big\{(t,t') \in T_i^{\text{dep}} \times T_j^{\text{dep}} : |t'-t| \le \delta, \ \ \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| < \varepsilon_h, \\ |z_i(t) - z_j(t')| < \varepsilon_z, \ \ D_{ij}(t,t') = 1\Big\},\end{aligned} \tag{2}$$

where the heading condition is active only when both pilots glide:

$$D_{ij}(t,t') = \begin{cases} \mathbf{1}\big[\hat{\mathbf{v}}_i(t) \cdot \hat{\mathbf{v}}_j(t') > \cos\theta_{\max}\big] & \text{if } s_i(t) = s_j(t') = \text{transition}, \\ 1 & \text{otherwise.} \end{cases} \tag{3}$$

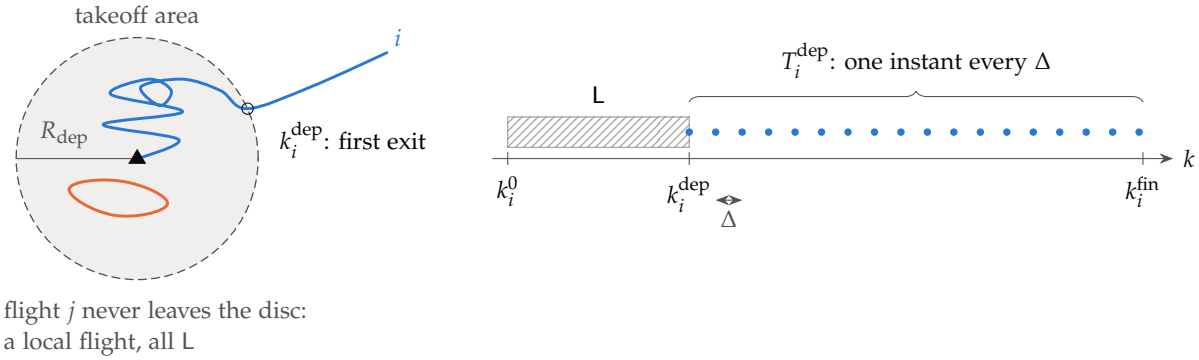



**Figure 6:** Local phase. Left: near the takeoff the pilot ridge soars and circles inside the disc of radius $R_{\text{dep}}$; the departure $k_i^{\text{dep}}$ is the first fix outside it. A flight that never leaves the disc is local. Right: the same flight in time; before the departure the state is L, after it the flight is evaluated on the grid instants $T_i^{\text{dep}}$.

> **In words.** $\mathcal{M}_{ij}$ is a set of *pairs of instants*. The pair $(t, t')$ belongs to it when $j$, at its instant $t'$, is where $i$ is at its instant $t$: within $\varepsilon_h$ horizontally and $\varepsilon_z$ vertically, with the two instants at most $\delta$ apart, and heading the same way if both glide. In the $(t, t')$ plane of Figure 7 it is a region inside the band $|t' - t| \le \delta$.

**Definition 4.2** (contact)**.**

$$m_{ij}(t) = \mathbf{1}\big[\exists\, t' :\ (t, t') \in \mathcal{M}_{ij}\big], \quad m_{ji}(t') = \mathbf{1}\big[\exists\, t :\ (t, t') \in \mathcal{M}_{ij}\big], \quad a_{ij}(t) = m_{ij}(t) \vee m_{ji}(t). \tag{4}$$

The conditions in (2) are symmetric, so $\mathcal{M}_{ji}$ is $\mathcal{M}_{ij}$ with the coordinates swapped and $a_{ij} = a_{ji}$. The two halves $m_{ij}$ and $m_{ji}$ differ in general; Appendix A shows why, on a worked example.

> **In words.** $m_{ij}(t) = 1$ when, at instant $t$, pilot $i$ is at a place that $j$ visits within $\delta$, before or after; $m_{ji}(t) = 1$ when, at instant $t$, pilot $j$ is at a place that $i$ visits within $\delta$. Their union $a_{ij}(t)$ is the symmetric indicator of contact between $i$ and $j$ at time $t$: at least one of the two is on the path of the other, within the delay $\delta$.

For the distance alone, $m_{ij}(t) = 1$ is the validity of instant $t$ in the sense of Li et al. [15, §IV-A] (Section 2.2): the distance minimised over the delays is below $\varepsilon_h$ exactly when some delay brings it below. The vertical tolerance, the heading condition tied to the flight phase and the symmetry of $a_{ij}$ are ours; the sign of the delay remains available to tell who follows whom (Section 4.5).

### 4.3 The role of each term

**Delay.** With $t' = t$ the condition is simultaneous proximity; with $t' \neq t$ it recognises similar but shifted tracks, the case of (R2)–(R3). If $t' > t$, $j$ passes where $i$ was, later: $j$ follows $i$. In Figure 7 all of $\mathcal{M}_{ij}$ lies above the diagonal. The contact asks only that some delay exists; how many seconds $j$ is behind $i$ is another question, taken up in Section 4.5.

**Tolerances: a box, not a ball.** $\varepsilon_h$ translates the "nearby places" of (R3): the tracks need not pass through the same points. Height gets a tolerance of its own because a group is flat: horizontally it spreads over scales of a kilometre, vertically it overlaps within a few hundred

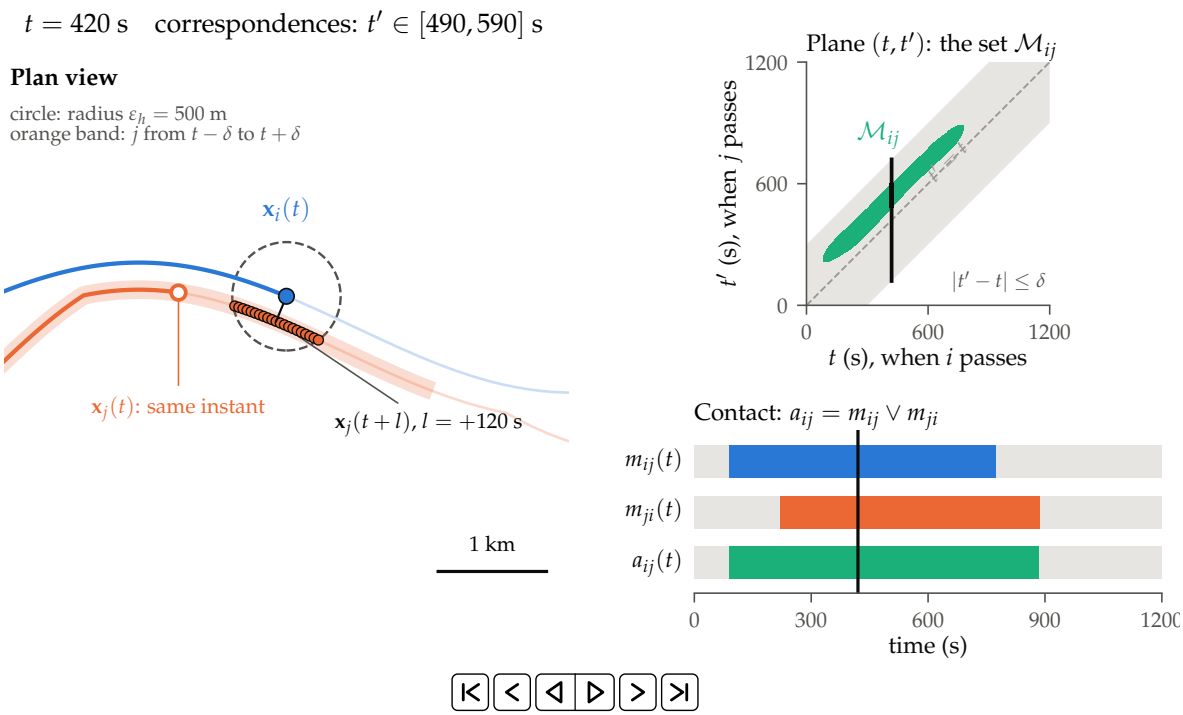



**Figure 7:** ▶ Correspondences and contact for a synthetic pair; the animation runs the instant $t$ from 0 to 1200 s. Left: plan view around $i$ at $t = 420$ s. Pilot $j$ flies the route of $i$ 120 s later and 250 m to the side; the circle has radius $\varepsilon_h$; the orange band is the part of the track of $j$ flown between $t - \delta$ and $t + \delta$; the dark dots are the instants $t'$ with $(t, t') \in \mathcal{M}_{ij}$, and the segment joins $\mathbf{x}_i(t)$ to the closest of them, at delay $l = +120$ s. At the same instant $j$ is about 1 km away: there is no simultaneous proximity. Top right: $\mathcal{M}_{ij}$ (green) in the $(t, t')$ plane, inside the band $|t' - t| \le \delta$ (grey); the black segment is the column of the instant of the left panel. $\mathcal{M}_{ij}$ lies above the diagonal because $j$ passes later. Bottom right: $m_{ij}$ projects $\mathcal{M}_{ij}$ on the $t$ axis, $m_{ji}$ on the $t'$ axis, and $a_{ij}$ is their union: $i$ is on the path of $j$ from 90 to 775 s, $j$ on the path of $i$ from 220 to 885 s. Model: $i$ flies at 8 m/s along $y = 600 \sin(2\pi s / 8\,\mathrm{km})$ m, where $s$ is the distance flown; $j$ joins the route at 300 s and leaves it at 800 s, drifting sideways at 3 m/s; altitude and heading conditions hold throughout.

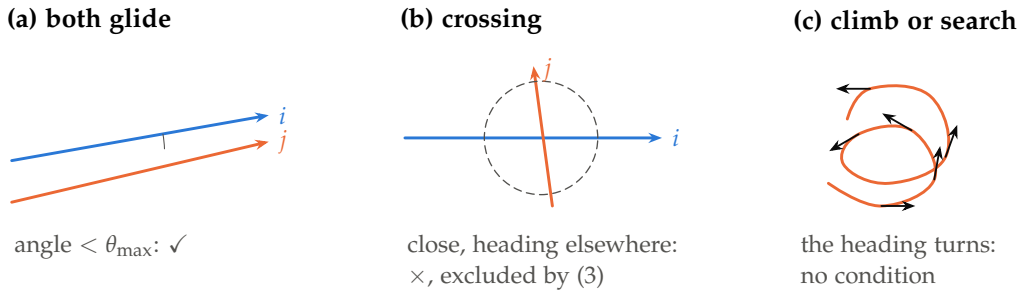



**Figure 8:** The heading condition (3). (a) Two gliders on the same route: their headings differ by less than $\theta_{\max}$. (b) A glider that crosses the area of $i$ heading elsewhere is close, and the condition excludes it. (c) In a climb or a search the heading turns continuously and says nothing about the route: the condition is off.

metres. A single $\varepsilon$ in $\|\Delta\mathbf{x}\|_{3D} < \varepsilon$ either makes altitude irrelevant, if it is of the order of a kilometre, or penalises the normal horizontal spread in glide, if it is of the order of 300 m (Figure 9). In a thermal the height is the only condition that acts: the pilots circle over the same spot, with a typical spiral radius of 20–26 m [1], so the horizontal distance stays far below $\varepsilon_h$, and only $\varepsilon_z$ separates pilots circling at different heights. Since $j$ climbs during the delay, the separation allowed there is $\varepsilon_z$ plus the height $j$ gains in $\delta$: $|z_i(t) - z_j(t)| < \varepsilon_z + w\delta$, 540 m at a climb rate $w = 2$ m/s with $\delta = 2$ min.

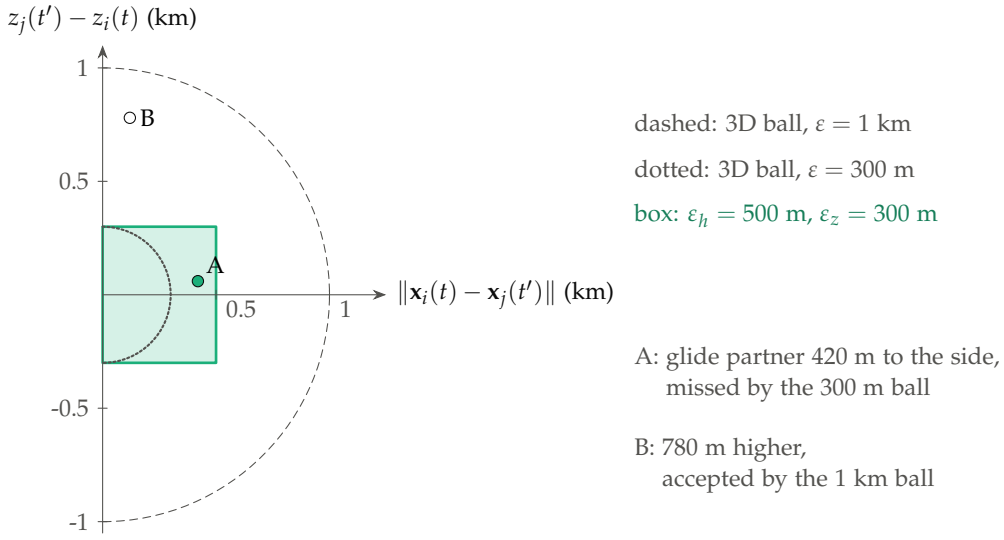



**Figure 9:** Tolerance regions for the separation between $(\mathbf{x}_i(t), z_i(t))$ and $(\mathbf{x}_j(t'), z_j(t'))$. The box of (2) is wide and flat, like a group: it keeps A and rejects B. A single 3D radius either rejects A or accepts B.

**Heading only in glide.** In glide the path is nearly straight and $\hat{\mathbf{v}}$ is the direction of the route: condition (3) excludes a pilot who crosses the same area heading elsewhere (Figure 8). In a climb the heading turns by $2\pi$ in about twenty seconds and carries no information on the route. In a search the pilot turns often, with partial circles and changes of heading, and the instantaneous heading says little about the route too: the condition is off whenever one of the two pilots is not in transition, unclassified fixes included. A rejected alternative is the velocity averaged over a window of a few minutes, with a minimum speed: in a long thermal without wind the mean velocity is nearly zero, and the threshold would remove the contacts exactly while two pilots share the thermal.

### 4.4 Choice of the maximum delay

If $j$ retraces the track of $i$ with a constant delay $\ell$, $\mathbf{x}_j(t) = \mathbf{x}_i(t - \ell)$, the distance between the two pilots at the same instant is the distance between two points of the track of $i$ flown $\ell$ apart,

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| = \|\mathbf{x}_i(t) - \mathbf{x}_i(t - \ell)\| \leq v_{\max}\,\ell,$$

and $v_{\mathrm{xc}}\,\ell$ on average once $\ell$ is longer than a climb–glide cycle, $v_{\mathrm{xc}}$ being the mean speed of advance. The distance at the same instant grows with the delay. Requiring that two pilots shifted by any delay up to $\delta$ stay within visual range means $v_{\mathrm{xc}}\,\delta \leq \varepsilon_{\mathrm{vis}}$ (Figure 10):

$$\delta \lesssim \frac{\varepsilon_{\mathrm{vis}}}{v_{\mathrm{xc}}}, \tag{5}$$

so $\delta \approx 5$ min for $\varepsilon_{\mathrm{vis}} \approx 1.5$ km and $v_{\mathrm{xc}} \approx 5$ m/s. The speed $v_{\mathrm{xc}}$ must be measured on the data, for example as the daily median of the speed of advance over 30-minute windows. The bound

gives the largest admissible delay, not the one at which following ends; on the data that one was measured directly and is 2 min (`method_assessment.pdf`).

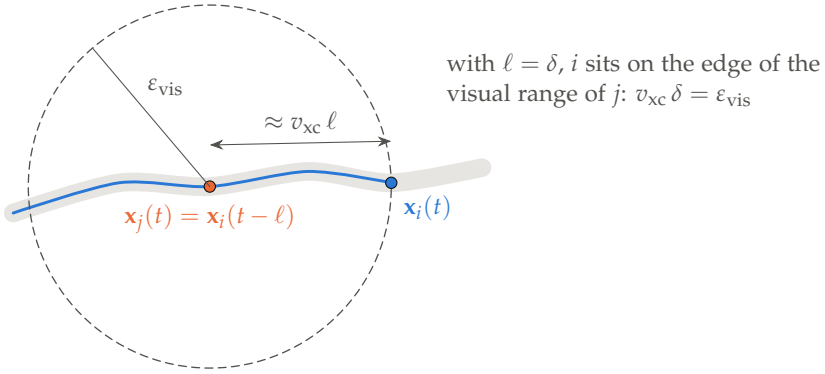



**Figure 10:** Choice of the maximum delay. A pilot that retraces the route of $i$ with delay $\ell$ is about $v_{\text{xc}}\ell$ behind; the dashed circle is the visual range around $j$. The largest delay that keeps $i$ in sight is $\delta = \varepsilon_{\text{vis}}/v_{\text{xc}}$.

*Remark* 4.3 (two notions of "together"). Flying together may mean *seeing each other*, and so being able to use social information, or *exploiting the same air masses*. The two notions can be written as indicators:

$$\begin{aligned} \text{visual co-presence:}\quad & V_{ij}(t) = \mathbf{1}\Big[\big\|\big(\mathbf{x}_i(t), z_i(t)\big) - \big(\mathbf{x}_j(t), z_j(t)\big)\big\| < \varepsilon_{\text{vis}}\Big], \\ \text{shared air:}\quad & m_{ij}(t) = \mathbf{1}\big[\exists\, |l| \le \delta :\ \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\| < \varepsilon_h, \\ & \qquad\qquad |z_i(t) - z_j(t+l)| < \varepsilon_z,\ D_{ij}(t, t+l) = 1\big]. \end{aligned}$$

Visual contact depends on the 3D distance and is simultaneous; exploiting the same thermals needs tight tolerances, vertical ones included, and admits a delay. Definition 4.2 with $\varepsilon_z$ of a few hundred metres is the second notion, restricted to the visual range by (5). Widening $\varepsilon_z$ brings the criterion closer to visual contact alone. The value of $\varepsilon_z$ is therefore a choice of definition, to be declared (Figure 11).

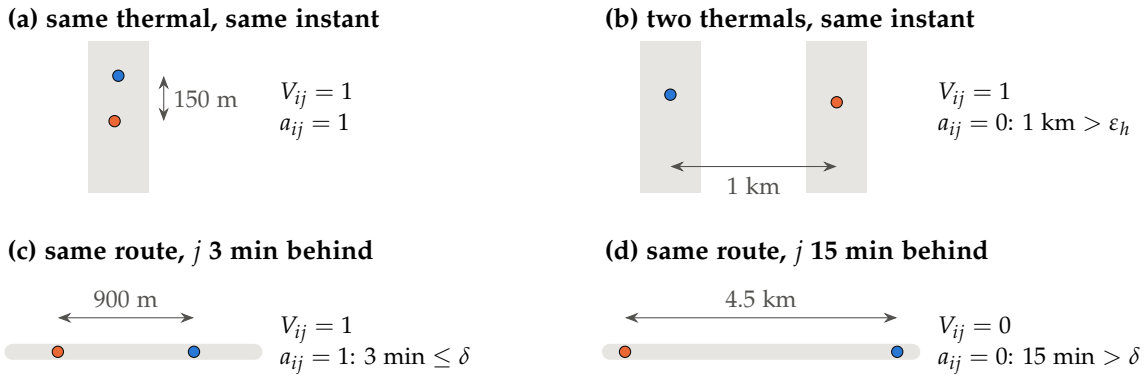



**Figure 11:** Visual co-presence $V_{ij}$ and contact $a_{ij}$ in four situations, with $\varepsilon_{\text{vis}} = 1.5$ km, $\delta = 5$ min and $v_{\text{xc}} = 5$ m/s. In (d), a delay of 15 minutes would put the two pilots in contact although they cannot see each other: this is what the bound (5) prevents.

## 4.5 How far behind: the delay of a contact

The contact asks only that some delay $|l| \le \delta$ exists, and the groups of Section 5 are built from it alone. To say who follows whom, for the comparison between followers and leaders

(Section 9), we also need its value: how many seconds $j$ is behind $i$. As in [15], at each instant $t$ with $m_{ij}(t) = 1$ we take the delay at which $j$ passes closest to where $i$ is,

$$l^*(t) = \arg \min_{l:\,(t,t+l)\in\mathcal{M}_{ij}} \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|, \tag{6}$$

and over a pair interval $[t_{\text{on}}, t_{\text{off}})$ (Definition 5.1) the signed median delay

$$\ell_{\text{med}} = \text{median}\{l^*(t) :\ t \in [t_{\text{on}}, t_{\text{off}}),\ m_{ij}(t) = 1\},$$

with $\ell_{\text{med}} > 0$ when $j$ follows $i$. How well $l^*$ is defined depends on the phase of flight.

**Glide.** The path is nearly straight and the distance has a single sharp minimum: $l^*$ is the delay at which $j$ crosses the point of $i$, +120 s in Figure 7.

**Thermal.** $j$ circles over the same spot, and the distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$, as a function of $l$, has a nearly equal minimum at every turn (Figure 12, top right): the minimiser is degenerate, and $l^*(t)$ falls anywhere among the accepted delays. Altitude narrows them. During the climb at rate $w$, pilot $j$ gains $wl$ metres in $l$ seconds:

$$z_j(t+l) \simeq z_j(t) + w\,l. \tag{7}$$

Write $\Delta z = z_i(t) - z_j(t)$ for how far $j$ is below $i$ at time $t$. The vertical condition then reads

$$|z_i(t) - z_j(t+l)| < \varepsilon_z \iff |\Delta z - w\,l| < \varepsilon_z \iff \Delta z - \varepsilon_z < w\,l < \Delta z + \varepsilon_z,$$

that is, dividing by $w > 0$,

$$l \in \left( \frac{\Delta z - \varepsilon_z}{w},\ \frac{\Delta z + \varepsilon_z}{w} \right), \tag{8}$$

an interval of width $2\varepsilon_z/w$ around $\Delta z/w$, the time $j$ needs to reach the altitude of $i$; the scan cuts it to $[-\delta, \delta]$. With the starting $\delta = 5$ min this at least halves the range of $l^*$. With the chosen $\delta = 2$ min the interval, 300 s wide at $w = 2$ m/s, is wider than $[-\delta, \delta]$ and altitude gains little: at $\Delta z = 200$ m, from $[-120, 120]$ to $(-50, 120]$ s.

**Search.** The pilot turns without climbing, $w \simeq 0$, and altitude constrains no delay. The instants at which either pilot is in search are left out of the median; $\ell_{\text{med}}$ is undefined when none remains.

# 5 From pairs to groups

## 5.1 Pair intervals

The instantaneous contact flickers wherever one of the quantities sits close to its threshold: a horizontal distance near $\varepsilon_h$ in glide, a height difference near $\varepsilon_z$, an angle near $\theta_{\max}$. It also flickers at the changes of state, when the heading condition switches on and off, and wherever the phase labelling is wrong on a few fixes.

**Definition 5.1** (pair interval)**.** For each pair consider the maximal runs of consecutive grid instants with $a_{ij} = 1$. Two runs separated by a gap shorter than $g$ are joined; the resulting runs shorter than $\tau^*$ are dropped. Each remaining run is a half-open interval $[t_{\text{on}}, t_{\text{off}})$, and the set of the intervals of the day is $\mathcal{I}$.

Joining the short gaps plays the role of the robustness of Buchin et al. [6, §4], which ignores interruptions shorter than $\alpha$; the maximal segments of Li et al. [15, 20] are an alternative. The threshold $\tau^*$ is fixed by the null model (Section 7).

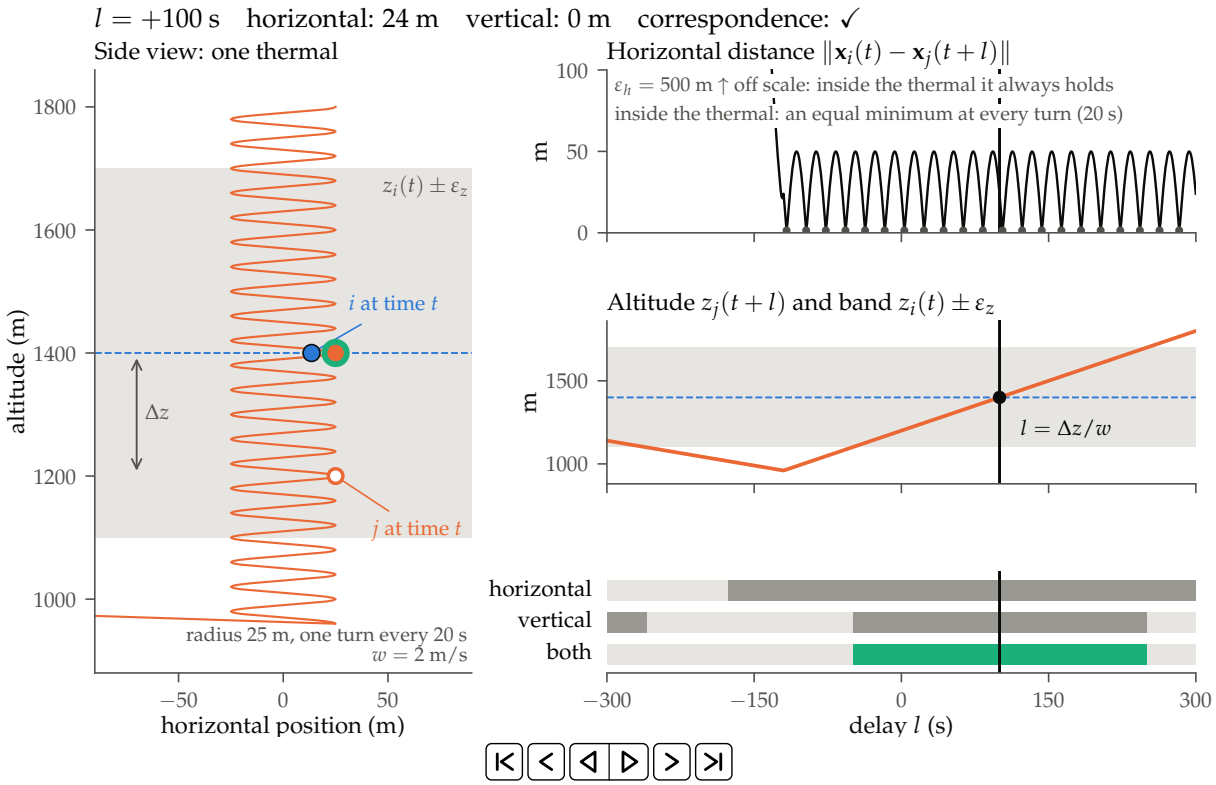



**Figure 12:** ▶ The delay in a thermal, for one instant $t$ and every delay $l$; the animation runs the delay $l$ from $-300$ to 300 s. Left: $i$ and $j$ in the same thermal, $j$ lower by $\Delta z = 200$ m; the band is $z_i(t) \pm \varepsilon_z$. Top right: the horizontal distance between $\mathbf{x}_i(t)$ and $\mathbf{x}_j(t+l)$; inside the thermal it never exceeds 50 m, far below $\varepsilon_h = 500$ m, so the horizontal condition alone accepts every delay of the stay; it has an equal minimum at every turn, so it cannot fix $l^*$. Middle right: the altitude of $j$ at $t+l$ crosses the band in a single window. Bottom: the delays that satisfy each condition and both; both hold for $-50 < l < 250$ s, the interval (8), centred on $\Delta z / w = 100$ s, and the scan keeps those with $|l| \le \delta$: $-50 < l \le 120$ s with $\delta = 2$ min. Before $l = -120$ s, $j$ was gliding towards the thermal: at $l \approx -280$ s it had the right altitude but was 1.4 km away. Model: $j$ circles with radius 25 m, one turn every 20 s, climbing at $w = 2$ m/s; it arrives at 9 m/s, sinking at 1 m/s; $i$ is on the same circle.

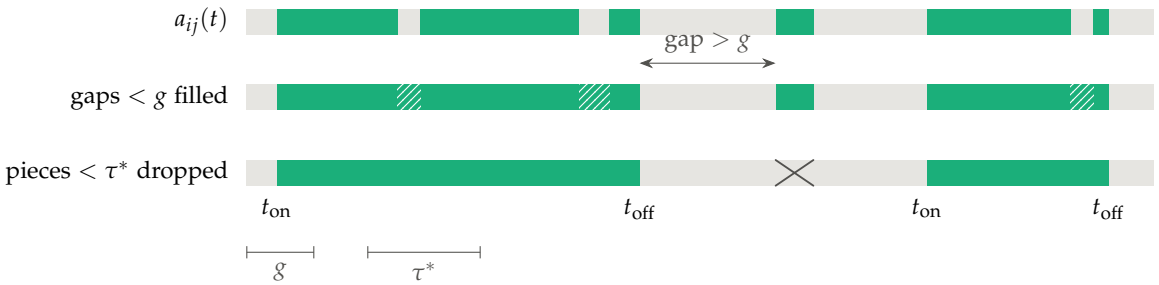



**Figure 13:** From the contact to the pair intervals. Top: the contact of one pair, with the flicker of the instantaneous condition. Middle: gaps shorter than $g$ are filled (hatched). Bottom: runs shorter than $\tau^*$ are dropped; two intervals remain.

## 5.2 The problem of non-transitivity

*Remark* 5.2. The relation "$i$ and $j$ are in contact" is symmetric but not transitive. If 1 flies with 8 in the first part of the afternoon and 8 flies with 14 later, 1 and 14 may never have met.

A partition into groups presupposes an *equivalence relation*, a relation $\sim$ that is reflexive ($i \sim i$), symmetric ($i \sim j \Rightarrow j \sim i$) and transitive ($i \sim j$ and $j \sim k \Rightarrow i \sim k$). Contact lacks the third property, so any partition must give up something (Figure 14):

(a) direct contact between all members: taking the transitive closure puts 1 and 14 in the same group although they never met;

(b) the partition: groups may overlap, here $\{1, 8\}$ and $\{8, 14\}$;

(c) the static picture: the groups are defined instant by instant.

We adopt (c): at every instant the groups are the connected components of the graph of active contacts, as the components of Buchin et al. [6], and the evolution in time is described without giving the groups a persistent identity. Their maximal groups, which may overlap, are a form of (b).

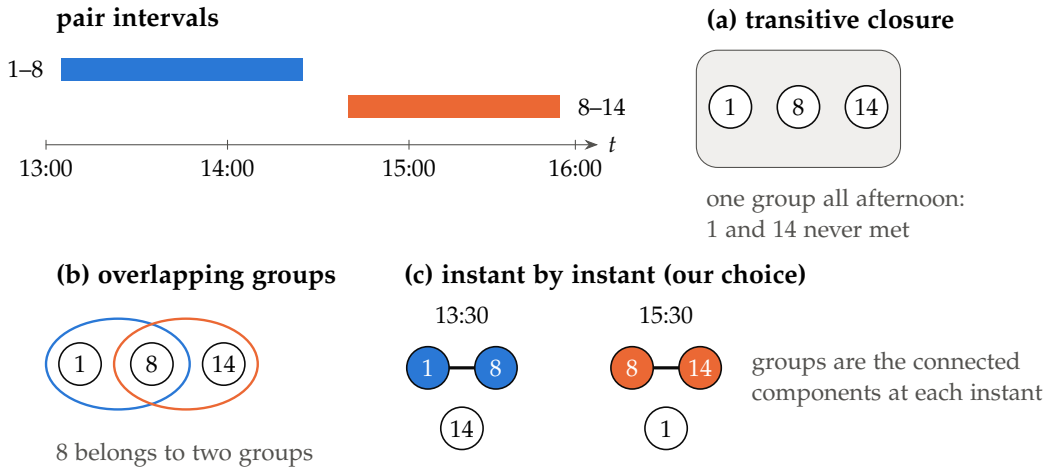



**Figure 14:** Non-transitivity. Flights 1 and 8 are in contact early in the afternoon, 8 and 14 later; 1 and 14 never meet. The three ways out: (a) one group containing all three, (b) two overlapping groups, (c) the groups of each instant, adopted here.

## 5.3 Configurations

The active graph at instant $t$ is

$$G(t) = \big(V(t), E(t)\big), \qquad V(t) = \{i : \ t \in T_i^{\text{dep}}\},$$
$$E(t) = \big\{\{i, j\} : \ \exists\, (i, j, t_{\text{on}}, t_{\text{off}}) \in \mathcal{I}, \ t \in [t_{\text{on}}, t_{\text{off}})\big\}. \tag{9}$$

> **In words.** $V(t)$ are the pilots that at instant $t$ have already departed, have not landed yet and have a position; $E(t)$ are the pairs whose contact interval contains $t$. A connected component of $G(t)$ is a group at instant $t$: its pilots are joined to each other by chains of contacts.

**Definition 5.3** (configuration)**.** A configuration $c$ is given by a set of flights $M_c$, its members, with $|M_c| \geq 2$, and by an interval of time $[t_c^{\text{start}}, t_c^{\text{end}})$, such that $M_c$ is a connected component of $G(t)$ at every $t$ of the interval and the interval is maximal. We write $c = \big(M_c, [t_c^{\text{start}}, t_c^{\text{end}})\big)$.

**Inside a configuration the members are fixed.**

Any change of members, one pilot joining or one leaving, ends the configuration and starts another. What can change inside a configuration is only the set of edges among its members.

The configurations are the edges of the Reeb graph of [6] with at least two entities, built on the graph of delayed contacts instead of simultaneous proximity. An edge that appears or disappears inside a component without changing its members, for example closing a triangle, does not create a new configuration (Figure 15).

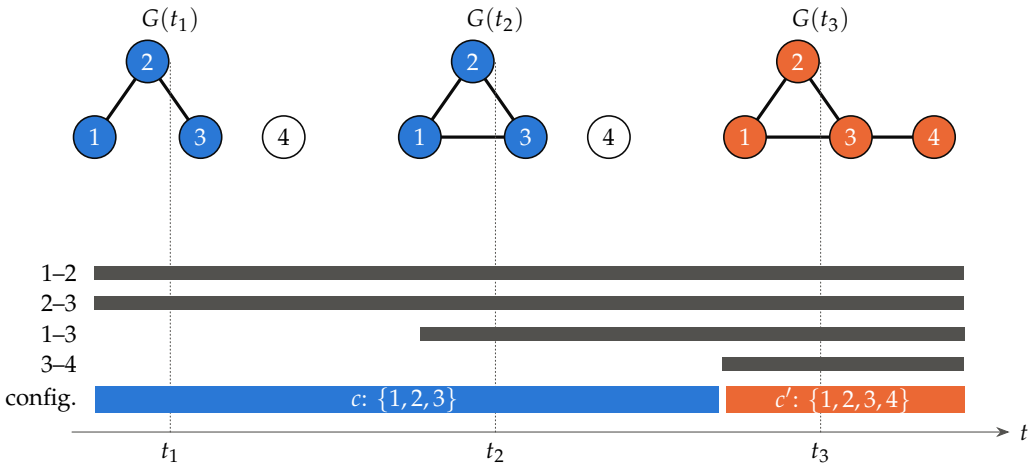



**Figure 15:** Configurations. Top: the active graph at three instants. Bottom: the pair intervals of the four flights and the configurations they produce. At $t_2$ the edge 1–3 closes a triangle: the component keeps its members and $c$ goes on. When 4 joins, $c$ ends and $c'$ starts.

**Proposition 5.4.** *At every instant each flight belongs to at most one configuration.*

*Proof.* The connected components of $G(t)$ form a partition of $V(t)$. □

The membership of flight $i$ at instant $t$ is therefore well defined:

$$\gamma_i(t) = \begin{cases} \mathsf{L} & \text{if } k_i^0 \le t < k_i^{\text{dep}} \text{ (local phase), or the flight is local,} \\ c & \text{if } i \in M_c \text{ and } t \in [t_c^{\text{start}}, t_c^{\text{end}}), \\ \mathsf{S} & \text{if } t \in T_i^{\text{dep}} \text{ otherwise (solo flight).} \end{cases} \tag{10}$$

In the tables, every instant between the departure and the landing that is not in a configuration is labelled S, including the instants in a gap of the track, where the flight has no position. The analyses that read the fixes see nothing there, so only the time fractions computed from the membership include them.

## 5.4 Episodes

**Definition 5.5** (episode)**.** Two configurations are linked, $c \to c'$, if one starts exactly when the other ends and they have at least one member in common:

$$c \to c' \iff t_{c'}^{\text{start}} = t_c^{\text{end}} \ \wedge \ M_c \cap M_{c'} \neq \emptyset. \tag{11}$$

An episode is a connected component of the undirected graph associated with $\to$.

**In words.** An episode is a chain of configurations without interruption: each one starts at the instant the previous one ends and keeps at least one of its members. Two configurations of the same episode need not share anyone; the chain joins them.

In the language of [6], an episode is a connected component of the Reeb graph after removing the edges of single pilots. Four properties matter (Figure 16).

(i) A split produces two configurations linked to the same previous one: they stay in the same episode. The same holds for a merge.

(ii) If the group dissolves completely, the chain breaks and the episode ends; a group that forms again later opens a new episode.

(iii) No threshold is needed: one member in common is enough.

(iv) The same flight can belong to several episodes of the same day, at different times.

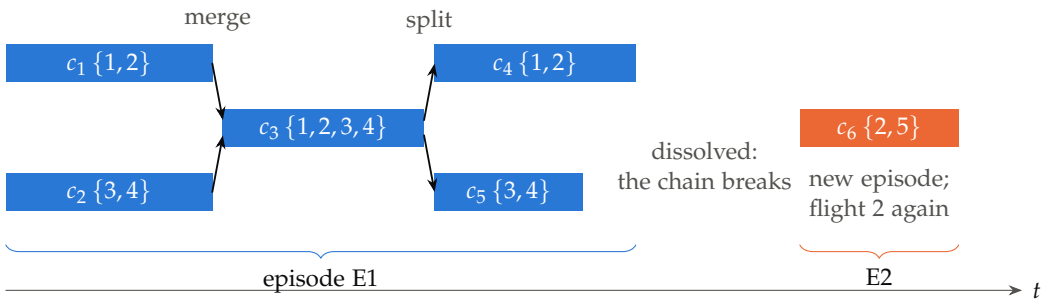



**Figure 16:** Episodes. Arrows are the links (11). The merge of $c_1$ and $c_2$ into $c_3$ and the split of $c_3$ into $c_4$ and $c_5$ keep all five configurations in episode E1, although $c_1$ and $c_5$ share no member. After the group dissolves, $c_6$ has no predecessor and opens E2: flight 2 belongs to both episodes.

*Remark* 5.6 (why not a persistent group name). One could keep the same group name when the overlap of the members between two instants exceeds a threshold, as methods that track communities in evolving networks do with the Jaccard index [9]. The choice has three defects: the threshold is arbitrary; if the members are replaced one at a time the group keeps its name even when none of the initial members is left; in splits and merges there is no natural criterion to decide who inherits the name. The episode does not claim to be "the same group": it records only the continuity, while the configuration records the composition.

## 5.5 Dependency blocks

**Definition 5.7** (block)**.** For each day consider the bipartite graph flights–episodes, with an edge between a flight and every episode it belongs to. Its connected components, restricted to the flights, are the *dependency blocks*; a flight never in a group is a block on its own.

**Proposition 5.8.** *The blocks are the finest partition of the flights in which the members of every episode are in the same block.*

*Proof.* A partition with this property must contain the transitive closure of the relation "share an episode", whose classes are exactly the connected components of the bipartite graph; these, in turn, satisfy the property. □

The blocks are the units resampled in the bootstrap (Section 9): flights of different blocks never shared an episode, flights of the same block did. Buchin et al. already note that a grouping structure splits the trajectories into independent data sets [6, §1].

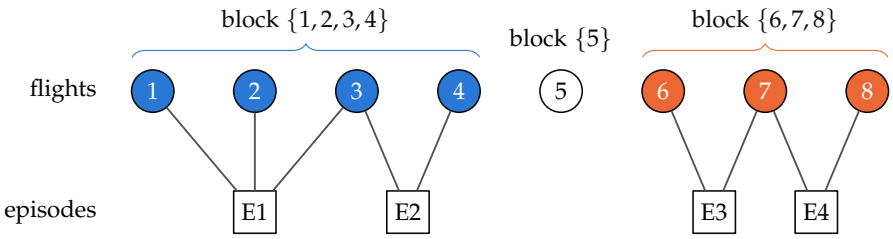



**Figure 17:** Dependency blocks of a day. Flight 3 is in episodes E1 and E2, so flights 1–4 form one block although 1 and 4 share no episode; flight 5 never flies in a group.

## 5.6 An example day

Figure 18 follows a day with five flights. At 11:40 flights 1, 8 and 22 meet (edges 1–8 and 8–22): configuration $c_1 = \{1, 8, 22\}$ starts. At 12:05 the edge 1–22 closes the triangle: the members do not change and $c_1$ goes on. At 12:55 the edge 1–8 closes and 1 changes route: $c_1$ ends and $c_2 = \{8, 22\}$ starts; they share 8 and 22, so $c_1 \to c_2$. At 13:30 the edge 8–14 opens: $c_3 = \{8, 14, 22\}$, and $c_2 \to c_3$. At 14:10 the group dissolves and episode E1 ends. At 15:00 flights 8 and 22 meet again: $c_4 = \{8, 22\}$ has no predecessor and opens episode E2.

Flights 1 and 14 belong to the same episode without ever being in the same configuration; 8 and 22 belong to two episodes; 1, 8, 14 and 22 form one dependency block, and flight 5, never in a group, a block of its own.

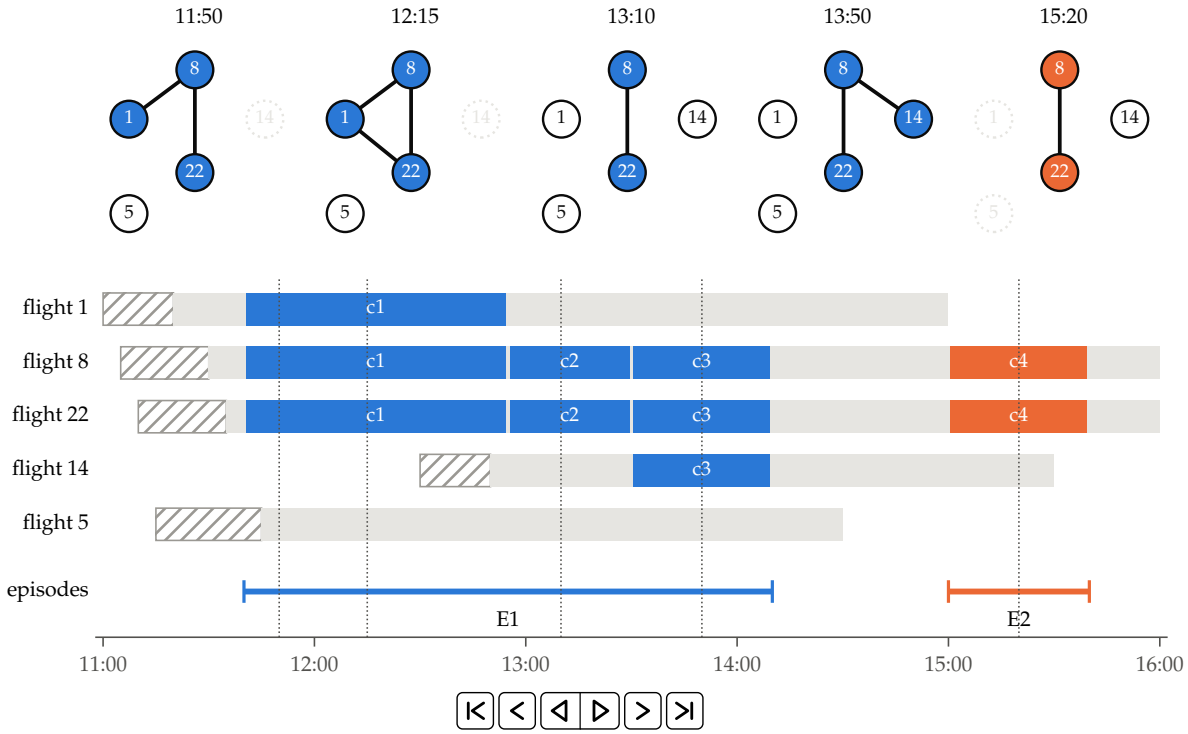



**Figure 18:** ▶ An example day; the animation runs the day from 11:00 to 16:00 and stops at each event. Top: the active graph $G(t)$ at the five dotted instants; filled nodes are in a configuration, coloured by episode; dashed nodes are in the local phase, faint ones not in flight. Bottom: the membership $\gamma_i(t)$ of each flight: hatched is the local phase L, grey is solo S, coloured segments are the configurations $c_1$–$c_4$, and the last row spans the episodes.

# 6 Algorithm

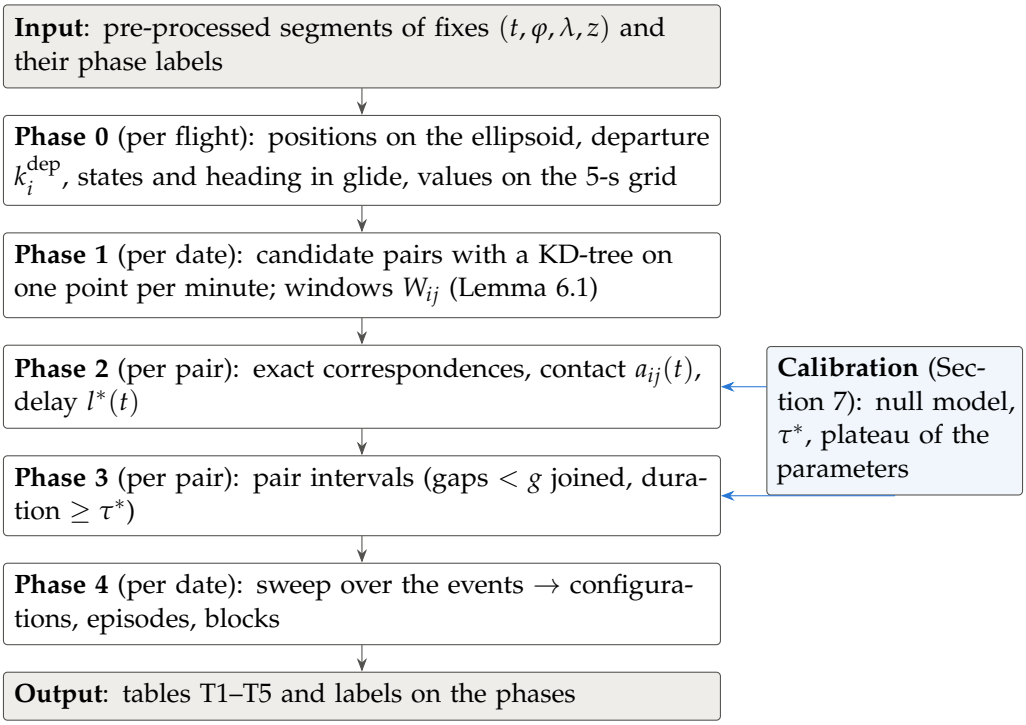



**Figure 19:** The method as a pipeline. Phases 1–4 work date by date: flights of different days do not interact, so the computation runs in parallel over the dates and needs the memory of one day.

## 6.1 Phase 0: preparation per flight

(a) **Positions.** The Earth-centred position of every fix (Section 3). One frame serves all the flights, at any distance and without a map projection, so the coordinates of two flights compare directly.

(b) **Altitude.** GNSS $z$, unchanged.

(c) **Departure.** $k_i^{\mathrm{dep}}$ from (1).

(d) **States and heading.** $s_i(k)$ from the phase labelling. In glide, $\hat{\mathbf{v}}_i(k)$ is the direction of the horizontal velocity, filtered with Savitzky–Golay, in the east and north directions of the first fix of the flight (the frame of the cleaned tracks), not of the fix itself. Two flights are then compared in frames whose axes differ by the rotation of the local north between their first fixes and the points compared: for a displacement $d$ to the east it is $(d/R)\tan\varphi$, about $3^\circ$ for 300 km at latitude $45^\circ$, small against $\theta_{\max} = 45^\circ$.

(e) **Grid.** The instants of $\Delta\mathbb{Z}$, with $\Delta = 5$ s, the same for all flights. A grid instant takes the position and the altitude of the fix at that instant or, between two fixes of the same segment at most $h_{\max} = 10$ s apart, the values interpolated linearly between them, with the state and the heading of the nearer fix; elsewhere the flight has no position. On a logger that records every second every grid instant is a fix. An interpolated position is at most $v_{\max}h/2$ from the true one, for two fixes $h$ apart: 69 m at 5 s, 139 m at 10 s.

**Is $\Delta = 5$ s enough in a thermal?**

Four points per turn do not draw the spiral (Figure 20), but the contact does not need its shape. What matters is how far a grid position can be from the track between two grid instants, compared with the tolerances:

- horizontally, a pilot is never more than $v_{\max}\Delta/2 \approx 69$ m from the nearest grid point in the worst case, about 20 m while circling at 8 m/s; $\varepsilon_h = 500$ m, and a whole thermal of radius 25 m fits in a small fraction of it;
- vertically, $w\Delta/2 = 5$ m at 2 m/s and at most 12.5 m at 5 m/s, against $\varepsilon_z = 300$ m;
- on the delay, a grid of 5 s against $\delta = 120$ s; in a thermal $l^*$ is undetermined over the whole window of accepted delays anyway (Section 4.5);
- on the ends of the intervals, $\pm\Delta$ against $g = 120$ s.

States and headings are computed on the fixes, before the grid, and lose nothing. The check of Section 7.3 repeats the computation with $\Delta = 1$ s on a subset of 1-Hz pairs.

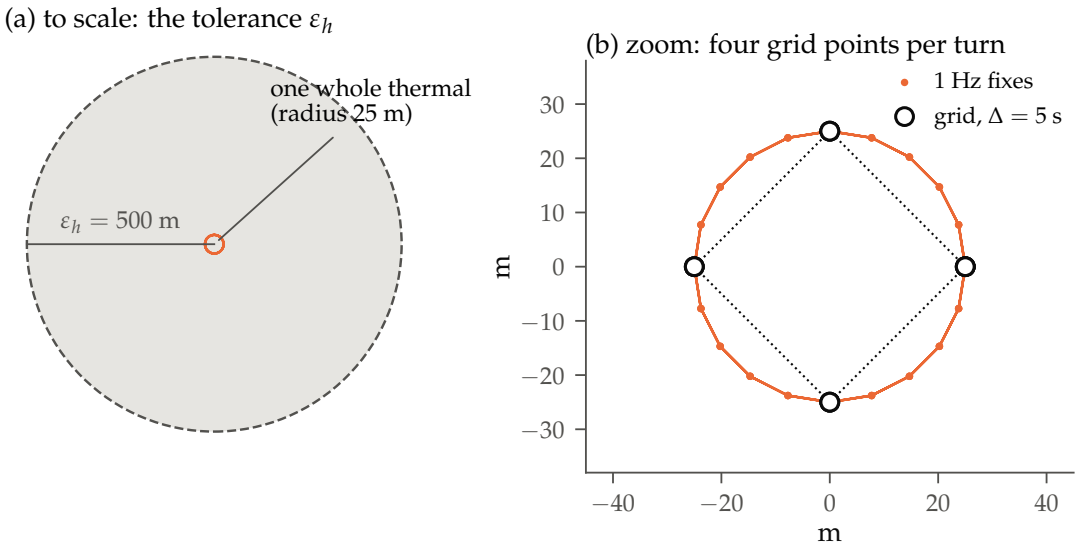



**Figure 20:** The 5-s grid in a thermal, to scale. (a) The horizontal tolerance $\varepsilon_h$ and a whole thermal of radius 25 m. (b) One turn of 20 s sampled at 1 Hz and on the grid of $\Delta = 5$ s: four grid points per turn, each within 20 m of the 1 Hz fixes around it.

| `date` | `flight` | `t` | `X, Y, Z` | `z` | `ve, vn` | `state` |
|---|---|---|---|---|---|---|
| partition | int32 | int32 (s) | float32 | float32 | float16 | int8 |

**Table 4:** Table of the grid points, in Parquet partitioned by date: about 30 bytes per row. `X, Y, Z` are the Earth-centred position, to 0.5 m in float32; `ve, vn` the unit heading in the east and north of the point.

Under the assumption, to be checked, of $1.86 \times 10^5$ flights lasting 1.5 h on average, the table has about $2 \times 10^8$ rows: 6 GB uncompressed, less in Parquet. It is worth keeping, because the null model and the calibration read it several times.

## 6.2 Phase 1: candidate pairs

The exact computation over all pairs of simultaneous flights is wasted effort: almost all of them never come close. A coarse filter reduces the pairs, with the guarantee of losing none.

The filter works on segments. Let $\tilde{\mathbf{x}}_i(s)$ be a segment of flight $i$ extended by holding its first position before it and its last position after it, and let $v_{\max}$ be the top speed of the pre-processing: within a segment no step between two fixes is faster, because the excursion rule (step 6) deletes or splits at any faster step, and the interpolation onto the grid keeps the bound. The filter compares positions at whole minutes, and a minute may fall just before or just after a segment: the extension gives the track a position there.

**Lemma 6.1** (no false negative). *If $(t, t') \in \mathcal{M}_{ij}$, $\tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_j$ are the extended segments that contain $t$ and $t'$, and $t_c = 60 \cdot \mathrm{round}(t/60)$, then*

$$\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| \le D_c := \varepsilon_h + v_{\max}\,(\delta + 60\ \mathrm{s}). \tag{12}$$

> **In words.** $t_c$ is the whole minute nearest to $t$ (Figure 21a), and both positions in (12) are read at that *same* instant. If two pilots have a correspondence at all, then at the nearest whole minute they are less than $D_c$ apart; a pair farther than $D_c$ at every minute has no correspondence and can be discarded.

*Proof.* The extension with a fixed position keeps the Lipschitz property $\|\tilde{\mathbf{x}}(s) - \tilde{\mathbf{x}}(s')\| \le v_{\max}|s - s'|$ for all $s, s'$. By construction of $t_c$, $|t - t_c| \le 30$ s, and $|t' - t_c| \le |t' - t| + |t - t_c| \le \delta + 30$ s. The triangle inequality along the path $\tilde{\mathbf{x}}_i(t_c) \to \mathbf{x}_i(t) \to \mathbf{x}_j(t') \to \tilde{\mathbf{x}}_j(t_c)$ (Figure 21) gives

$$\begin{aligned}\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| &\le \|\tilde{\mathbf{x}}_i(t_c) - \mathbf{x}_i(t)\| + \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| + \|\mathbf{x}_j(t') - \tilde{\mathbf{x}}_j(t_c)\| \\ &\le 30\, v_{\max} + \varepsilon_h + v_{\max}(\delta + 30),\end{aligned}$$

where the middle term is below $\varepsilon_h$ because $(t, t')$ is a correspondence (Definition 4.1). □

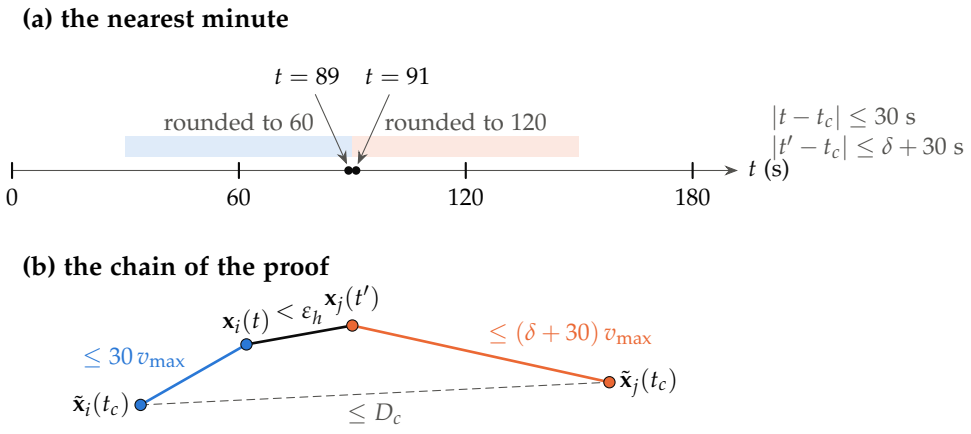



**Figure 21:** Lemma 6.1. (a) Every instant is at most 30 s from its nearest whole minute. (b) From $i$ at the minute to $j$ at the same minute through the correspondence $(t, t')$: each step is bounded by the time it takes at most $v_{\max}$, or by $\varepsilon_h$.

With $v_{\max} = 100$ km/h, $\delta = 300$ s and $\varepsilon_h = 500$ m, $D_c \approx 10.5$ km; with the delay chosen on the data, $\delta = 120$ s, $D_c \approx 5.5$ km. The filter runs as follows, for each date.

1. For each segment keep one point per minute of its extended track, for $t_c$ from $\delta + 60$ s before its first instant after the departure to $\delta + 60$ s after its last instant. This is the interval in which the lemma may ask for the position, whichever of the two roles the flight has.

2. Build a single KD-tree [25] on the coordinates $(X, Y, Z, Kt_c/60)$ with $K \gg D_c$, for example $K = 10^8$ m: points of different minutes are more than $D_c$ apart and are never paired (Figure 22).

3. The search of all pairs within $D_c$ (`query_pairs`) returns pairs of points of the same minute, hence triples $(i, j, t_c)$; the pairs of two segments of the same flight are dropped.

4. For each candidate pair, with $i < j$, the window is $W_{ij} = \bigcup_{t_c} [t_c - 30, t_c + 30] \cap T_i^{\text{dep}}$, the union over the minutes $t_c$ at which the pair passed the filter: a set of instants of the flight of $i$.

By the lemma, every correspondence $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$. The cost is $O(n_d \log n_d)$ per date, with $n_d$ the number of points per minute of the day. Phase 1 is worth running on the whole dataset before Phase 2: the number of pairs and the total duration of the windows give a precise estimate of the cost that follows.

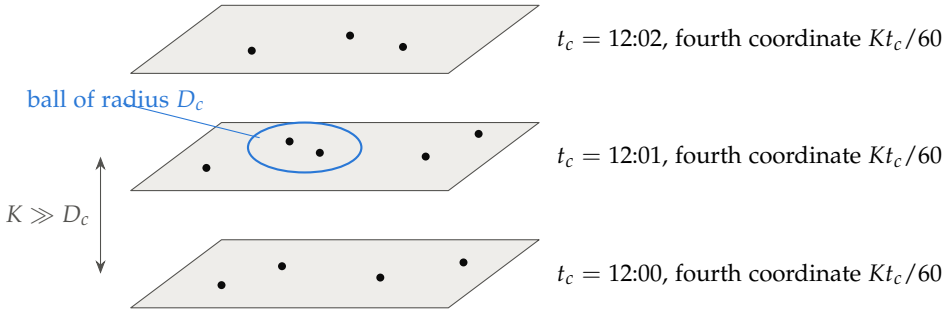



**Figure 22:** The KD-tree of Phase 1. Each minute is a layer of the space $(X, Y, Z, Kt_c/60)$, drawn here as a plane, and the layers are $K \gg D_c$ apart: a search of radius $D_c$ only finds points of the same minute, so one tree does the work of one search per minute.

### 6.3 Phase 2: exact correspondences

**Lemma 6.2.** *Scanning the instants $t \in W_{ij}$ and the delays $l \in \Lambda = \{-L\Delta, \ldots, L\Delta\}$, with $L = \lfloor \delta/\Delta \rfloor$, enumerates all pairs of $\mathcal{M}_{ij}$; hence $m_{ij}$ and $m_{ji}$ are obtained exactly.*

*Proof.* Every $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$ (Lemma 6.1) and $t' - t \in \Lambda$. Moreover $m_{ji}(t') = 1$ if and only if there is a $t$ with $(t, t') \in \mathcal{M}_{ij}$, and every such $t$ lies in $W_{ij}$. □

Each pass of the loop takes one delay $l$: in the $(t, t')$ plane it is the diagonal $t' = t + l$, evaluated over the whole window at once (Figure 23). The cost is $\sum_{\text{pairs}} |W_{ij}| \, (2L + 1)$ distance evaluations, with $2L + 1 = 121$ for $\delta = 5$ min and $\Delta = 5$ s, 49 for $\delta = 2$ min. On the whole archive (6 209 dates) the run takes about an hour with five processes. The memory is $O(|W_{ij}|)$ per pair: boolean arrays are accumulated, without generating the list of pairs of points. Each step of the loop is a vector operation on arrays.

**Algorithm 1** Phase 2: scan of one candidate pair

**Input:** flights $i < j$, on the grid; window $W \subseteq T_i^{\text{dep}}$; delays $\Lambda$
**Output:** contact $a$ on the support $W \cup (W + \Lambda)$; delay $l^*$ on $W$

$m_{ij} \leftarrow 0$; $m_{ji} \leftarrow 0$; $d^{\min} \leftarrow +\infty$; $l^* \leftarrow$ undefined
**for** $l \in \Lambda$ **do**
  $\tau \leftarrow \{t \in W : \ t + l \in T_j^{\text{dep}}\}$ ▷ array of instants
  $d \leftarrow \|\mathbf{x}_i(\tau) - \mathbf{x}_j(\tau + l)\|$; $h \leftarrow |z_i(\tau) - z_j(\tau + l)|$
  $c \leftarrow (d < \varepsilon_h) \wedge (h < \varepsilon_z) \wedge D_{ij}(\tau, \tau + l)$
  $m_{ij}[\tau] \leftarrow m_{ij}[\tau] \vee c$; $m_{ji}[\tau + l] \leftarrow m_{ji}[\tau + l] \vee c$
  $u \leftarrow c \wedge (d < d^{\min}[\tau])$; $d^{\min}[\tau] \leftarrow d$ and $l^*[\tau] \leftarrow l$ where $u$
$a(t) \leftarrow m_{ij}(t) \vee m_{ji}(t)$
**return** $a$, $l^*$

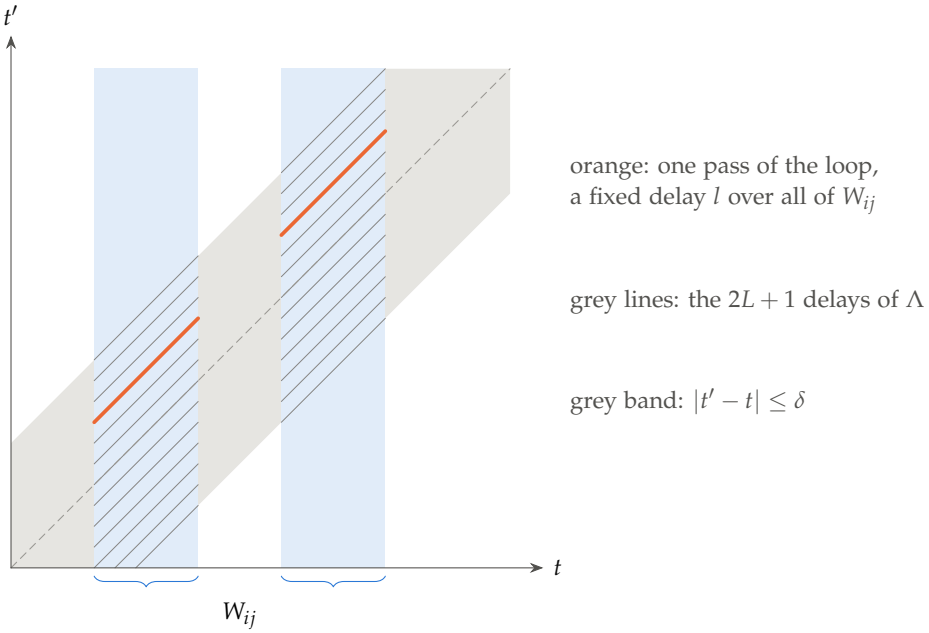



**Figure 23:** Phase 2 in the $(t, t')$ plane. Only the columns of the window $W_{ij}$ are scanned, one diagonal $t' = t + l$ at a time; by Lemma 6.2 this covers all of $\mathcal{M}_{ij}$.

### 6.4 Phase 3: pair intervals

From the contact $a$ one extracts the runs of 1, joins those separated by gaps shorter than $g$ and keeps those lasting at least $\tau^*$ (Definition 5.1). The contact is first restricted to the time from the later of the two departures to the earlier of the two landings, so that every edge of the graph (9) joins two nodes of $V(t)$. This cuts at most $\delta$ at each end of a contact, where $m_{ji}$ could hold at an instant at which $i$ has not yet departed or has already landed. For each interval one also records:

- the signed median delay $\ell_{\text{med}}$, which tells who follows whom and does not enter Phase 4 (Section 4.5);
- the distance between the takeoffs $d_{\text{dec}} = \|\mathbf{x}_i(k_i^0) - \mathbf{x}_j(k_j^0)\|$, which allows requirement (R1) to be imposed afterwards, if wanted.

### 6.5 Phase 4: configurations and episodes

Between one event and the next the set of active edges does not change, so it is enough to update the components at the instants at which an interval starts or ends, and only around the nodes involved (Figure 24).

**Algorithm 2** Phase 4: sweep over the events of one day

**Input:** pair intervals $\mathcal{I}$ of the day
**Output:** configurations; episodes (classes of a union–find)

events $\leftarrow \{(t_{\text{on}}, +, \{a,b\}),\ (t_{\text{off}}, -, \{a,b\})\}$ sorted in time
$A \leftarrow \emptyset$ (active edges); $\mathcal{O} \leftarrow \emptyset$ (open configurations)
**for** each instant $t_e$, with its batch of events $B$ **do**
  $U \leftarrow$ endpoints of the edges in $B$ ▷ nodes involved
  $R \leftarrow U \cup \bigcup \{M_c :\ c \in \mathcal{O},\ M_c \cap U \neq \emptyset\}$
  update $A$ with the events of $B$
  $\mathcal{K} \leftarrow$ connected components of $(R, A|_R)$ with at least two nodes ▷ local visit
  $X \leftarrow \{c \in \mathcal{O} :\ M_c \subseteq R,\ M_c \notin \mathcal{K}\}$; close every $c \in X$ with $t_c^{\text{end}} \leftarrow t_e$
  **for** $K \in \mathcal{K}$ that is not already an open configuration **do**
    open $c'$ with $M_{c'} = K$ and $t_{c'}^{\text{start}} = t_e$
    **for** $c \in X$ with $M_c \cap K \neq \emptyset$ **do**
      Union$(c, c')$

**Lemma 6.3.** *After the update, the connected component of every node of $R$ is contained in $R$; hence the local recomputation of Algorithm 2 gives the same components as a global one.*

*Proof.* Before the update the open configurations are disjoint, and those that touch $U$ lie entirely in $R$. A path that starts in $R$ uses old edges, which stay inside the old component, or new edges, whose endpoints lie in $U \subseteq R$. By induction on the length of the path, every node it reaches is in $R$. □

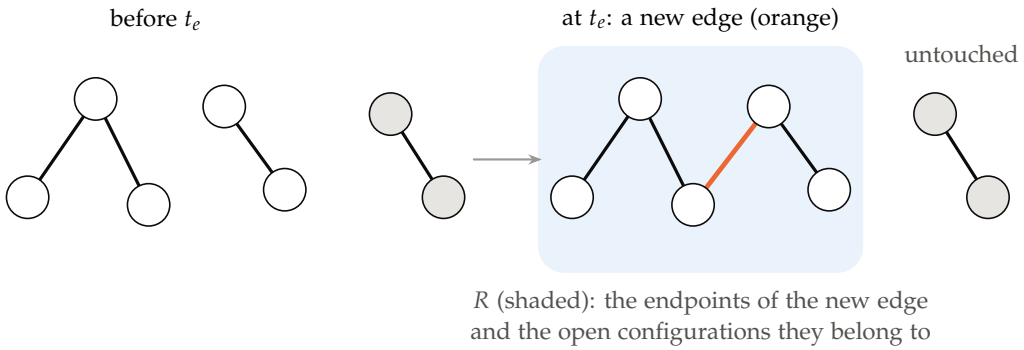



**Figure 24:** One step of the sweep. Only the nodes of $R$ are visited again; the grey component far away keeps its configuration. The animation of Figure 18 runs the sweep over a whole day.

The cost per day is $O(|\mathcal{I}| \log |\mathcal{I}|)$ for the sorting, plus the sum of the sizes of the components involved at each event, at most a few hundred nodes. The classes of the union–find [24] are the episodes. From the list of the configurations one obtains $\gamma_i(t)$ as in (10), filling with S the stretches after the departure not covered by a configuration and with L the local phase.

| Phase | Cost | Data produced |
|---|---|---|
| 0 | linear in the number of fixes ($\sim 10^9$) | grid points, $\sim 2 \times 10^8$ rows |
| 1 | $O(n_d \log n_d)$ per date, $\sum_d n_d \sim 1.7 \times 10^7$ | candidate pairs and windows |
| 2 | $\sum \lvert W_{ij} \rvert (2L+1)$; measured after Phase 1 | temporary arrays per pair |
| 3 | linear in the duration of the windows | T1, pair intervals |
| 4 | $O(\lvert \mathcal{I} \rvert \log \lvert \mathcal{I} \rvert)$ per date | T2–T5 |

**Table 5:** Costs and space. The numbers assume $1.86 \times 10^5$ flights of 1.5 h on average and are to be checked on the data.

### 6.6 Costs and space

Phases 1–4 are independent across dates and run in parallel without coordination. Phase 2 dominates the cost; Phase 1 allows it to be estimated before running it.

## 7 Calibration and validation

### 7.1 Null model and minimum duration

Two independent pilots on the same route can be in contact by chance, for short stretches. The null model estimates how often this happens under the same terrain and weather, without interaction. Interaction indices based on standard tests overestimate the interactions with finely sampled telemetry [16], so the threshold is calibrated empirically. The idea of shifting tracks in time, which keeps the spatial constraints and destroys the synchronisation, comes from Spiegel et al. [22], who shuffle whole days of each individual; here a flight is shifted within its day, against one other flight.

1. **Surrogate pairs.** Choose pairs $(i, j)$ of the same day with takeoffs within a radius $R_s$ (for example 10 km) and replace $j$ with its copy shifted in time:

$$\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r_{ij}), \quad z_j^s(t) = z_j(t + r_{ij}), \qquad r_{ij} = (k_j^0 - k_i^0) - \vartheta, \quad \vartheta \sim \hat{P}\big(k_j^0 - k_i^0\big). \tag{13}$$

   The copy of $j$ takes off at $k_i^0 + \vartheta$. The variable $\vartheta$ is drawn from the empirical distribution of the shifts between the takeoffs of the real candidate pairs of Phase 1, so that the shift of the surrogate pair is realistic. Only shifts with $|r_{ij}| \geq r_{\min}$ are accepted, with $r_{\min} > 2\delta$.

2. **Destruction of the real interactions.** If $(t, t')$ was a real correspondence, in the surrogate pair the position of $j$ at instant $t'$ appears at instant $t' - r_{ij}$, which is at least $|r_{ij}| - \delta > \delta$ away from $t$: no real correspondence survives (Figure 25). The correspondences that remain are chance encounters on the same route.

3. **Null durations.** Phases 2–3 run on the surrogate pairs without the threshold $\tau^*$, giving the durations of the chance intervals and the exposure $h_{ij}$, the duration of the overlap between $T_i^{\text{dep}}$ and the shifted $T_j^{\text{dep}}$.

4. **False-positive rate.** With $\rho_0(\tau)$ the number of surrogate intervals lasting at least $\tau$ per unit of exposure, $H$ the total exposure of the real pairs (same day, takeoffs within $R_s$) and $N(\tau)$ the number of real intervals lasting at least $\tau$:

$$\widehat{\mathrm{FDR}}(\tau) = \frac{\rho_0(\tau)\, H}{N(\tau)}, \qquad \tau^* = \min\big\{\tau : \ \widehat{\mathrm{FDR}}(\tau) \leq q\big\}, \qquad q = 0.05. \tag{14}$$

**In words.** $\rho_0(\tau)H$ is the number of intervals of duration at least $\tau$ that chance alone would produce among the real pairs. Dividing it by the number actually observed gives the share of chance intervals among them; $\tau^*$ is the shortest duration at which that share falls to 5% (Figure 26).

This is the estimate of the false discovery rate of SAM [26], the number of discoveries expected by chance over the number observed, with time shifts in place of the permutations.

The variability of $\tau^*$ is estimated by resampling the dates. Since $\vartheta$ follows the takeoff delays of the day, the shifts are of the order of an hour, and the copy flies in the weather of another hour (the cloud base rises, the thermals move): what the null model calls real then includes pilots who share a thermal at the same time. Shorter shifts would keep the weather, but would also bring within $\delta$ the pilots who follow each other a few minutes apart, and count them as chance. We keep the shifts that follow the takeoff delays; $\tau^*$ depends on this choice (`method_assessment.pdf`).

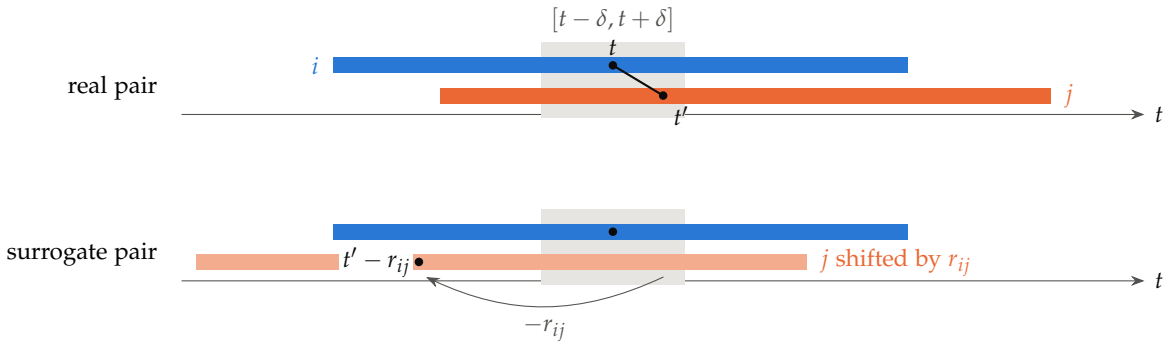



**Figure 25:** Null model. Top: a real correspondence $(t, t')$, with $t'$ inside the window $[t-\delta, t+\delta]$ (grey). Bottom: the track of $j$ shifted by $r_{ij}$ with $|r_{ij}| > 2\delta$; the same point of $j$ now comes at $t' - r_{ij}$, outside the window, so the real correspondence is gone and only chance encounters remain.

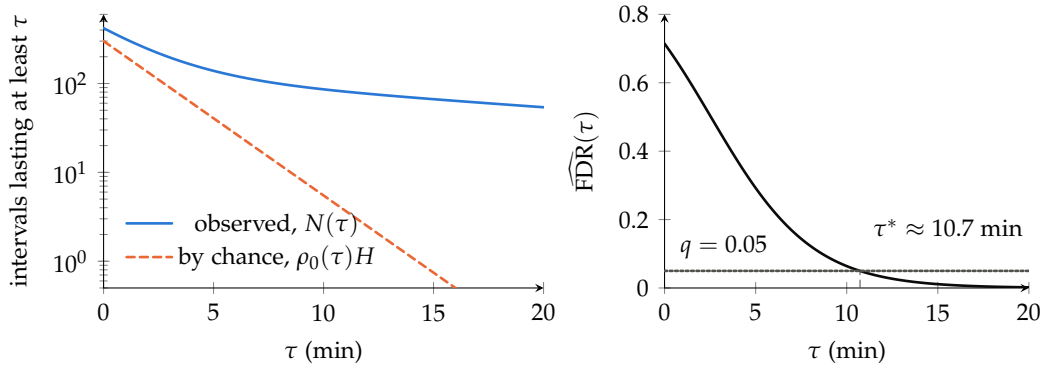



**Figure 26:** Choice of $\tau^*$, on schematic curves: $N(\tau) = 300\,e^{-\tau/2.5} + 120\,e^{-\tau/25}$ and $\rho_0(\tau)H = 300\,e^{-\tau/2.5}$, with $\tau$ in minutes. Chance contacts are many and short, real ones fewer and longer; $\tau^*$ is where the share of chance intervals falls to $q$.

## 7.2 Plateau of the parameters

Let $f_G$ be the fraction of the flight time after the departure spent in a group:

$$f_G = \frac{\sum_i \left|\{t \in T_i^{\mathrm{dep}} : \ \gamma_i(t) \notin \{\mathsf{L}, \mathsf{S}\}\}\right|}{\sum_i |T_i^{\mathrm{dep}}|}. \tag{15}$$

The parameters $(\varepsilon_h, \varepsilon_z, \theta_{\max}, \delta, g)$ are varied around their starting values, and the values are chosen in a zone where $f_G$ and the number of episodes are stable (Figure 27). If there is no plateau, the thresholds decide the result, and this must be said. On the data there is none (`method_assessment.pdf`).

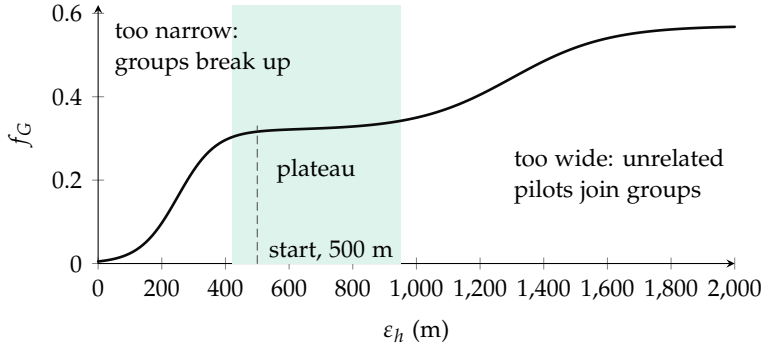



**Figure 27:** A plateau of the fraction in a group, on a schematic curve $f_G = 0.32/(1 + e^{-(\varepsilon_h - 250)/60}) + 0.25/(1 + e^{-(\varepsilon_h - 1300)/150})$, with $\varepsilon_h$ in metres. A starting value inside the flat zone makes the result insensitive to the exact threshold.

### 7.3 Other checks

- Visual inspection: about ten episodes on a map, with the entries and exits checked by hand.
- Sensitivity to the step: on a subset of pairs, the computation with $\Delta = 1$ s must give the same intervals.
- Competition days: if the dataset contains competition tracks, on those days everybody flies the same course and is in a group by construction; they are flagged and treated apart.
- Grid: the share of grid instants without a position, per flight and by logging interval.

## 8 Outputs

The state of flight $i$ at instant $t$ is obtained with an as-of join on (`flight`, $t$) in T3, sorted by `t_start` (Figure 28). T3 has about $N_{\text{flights}}(2 + 2\bar{c})$ rows, with $\bar{c}$ the mean number of configurations per flight: of the order of $10^5$–$10^6$ rows, tens of MB. The other tables are smaller.

## 9 Use for the comparison between group and solo flight

**Labels on the phases.** For a run $\sigma$ of one phase (climb, search or glide) of flight $i$, with interval $J_\sigma$, define the fraction in a group and the mean size:

$$f_\sigma = \frac{\left|\{t \in J_\sigma \cap T_i^{\text{dep}} : \gamma_i(t) \notin \{\mathsf{L}, \mathsf{S}\}\}\right|}{|J_\sigma \cap T_i^{\text{dep}}|}, \qquad \bar{n}_\sigma = \text{mean of } n_i(t) \text{ over } J_\sigma \cap T_i^{\text{dep}}, \tag{16}$$

with $n_i(t)$ the number of members of the configuration of $i$ at $t$, 1 if solo. The run is *in a group* if $f_\sigma \geq 1/2$ and *solo* if $f_\sigma = 0$; the intermediate cases are excluded, or $f_\sigma$ is used as a continuous covariate (Figure 29). Runs in the local phase are excluded.

| Table | Columns (type) and content |
|---|---|
| T1 `pair_intervals` | `date`, `a`, `b` (int32), `t_on`, `t_off` (int32, s), `duration` (int32), `ell_med` (int16, s, signed: positive if `b` follows `a`), `d_dec` (float32, m). Who was with whom, and when. |
| T2 `configurations` | `date`, `conf_id`, `episode_id` (int32), `t_start`, `t_end` (int32), `n` (int16), `members` (list of int32). |
| T3 `membership` | `flight` (int32), `t_start`, `t_end` (int32), `state` (int8: L, S, G), `conf_id`, `episode_id` (int32, null if L or S), `n` (int16, 1 if S). The main table: each flight cut into stretches of constant membership. |
| T4 `episodes` | `episode_id`, `date`, `t_start`, `t_end`, `n_flights`, `n_max`, `n_config`, `block`. |
| T5 `flight_summary` | `flight`, `phi_G` (fraction of $T_i^{\text{dep}}$ in a group), `kappa` (distinct companions), `n_entries`, `n_episodes`, `block`, `local` (boolean). |

**Table 6:** Final tables, in Parquet. The members of a configuration are not repeated in T3: they are read from T2 through `conf_id`.

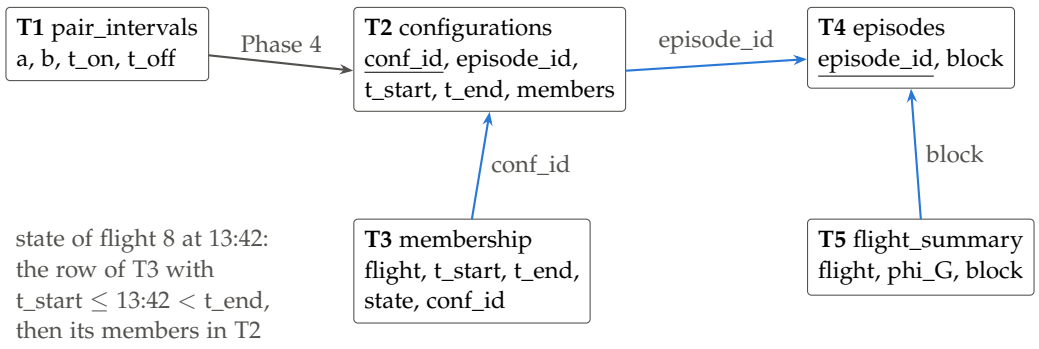



**Figure 28:** How the tables connect. Arrows point from a column to the table where it is a key. The example reads the state of a flight at one instant with an as-of join on T3.

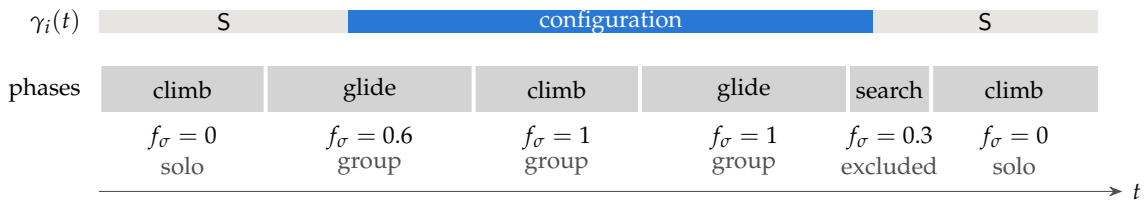



**Figure 29:** Labels on the phases of one flight. Each run of one phase gets the fraction $f_\sigma$ of its duration spent in a configuration: runs with $f_\sigma \geq 1/2$ count as group, those with $f_\sigma = 0$ as solo, the search with $f_\sigma = 0.3$ is left out.

**Stratification.** Group flights concentrate on the good days and the busy sites: a global comparison would mix the effect of the group with that of the weather. The comparison runs within strata $s$ (date and area):

$$\hat{\Delta} = \sum_{s} w_s \left( \hat{\theta}_{G,s} - \hat{\theta}_{S,s} \right), \tag{17}$$

where $\hat{\theta}$ is the statistic of interest and the weights $w_s$ are proportional, for example, to the number of group runs of the stratum.

**Uncertainties.** Runs of pilots of the same episode are not independent, and neither are those of the same flight. The bootstrap resamples the dependency blocks (Section 5.5) within the strata.

**Confounders.** Competition days; the level of the pilot (the same pilots often fly together); the glider class (EN A–D); the time of day.

**Questions to ask.** Distributions of the times spent in thermals and of the lengths of the glides, in a group and alone; mean squared displacement and effective exponent; differences between followers and leaders, by the sign of $\ell_{\text{med}}$. In birds, followers leave the thermals earlier and lower [8], and individuals with social information glide faster [28]: these are hypotheses to test on pilots before assuming them.

## 10 Choices to declare

The starting values and those chosen on the data are in Table 3. The thesis should state these choices, each argued in the section given or in the companion note `method_assessment.pdf`:

- $\varepsilon_z$ encodes the notion of "together": shared thermals if tight, visual contact if wide (Section 4.4);
- $\delta$ is bounded by the visual range through $v_{\text{xc}}$ (Section 4.4) and measured on the data at 2 min (`method_assessment.pdf`);
- $q = 0.1$: at most one contact in ten is chance (`method_assessment.pdf`);
- $\varepsilon_h$, $\varepsilon_z$ and $g$ show no plateau on the data, so their values are definitions (`method_assessment.pdf`);
- headings are compared only when both pilots glide; climb, search and unclassified fixes turn the condition off (Section 4.3); in the run reported here the phases are Vilpellet's, on 1 Hz flights only, so the condition acts only between two such flights (Section 3);
- contacts before the departure are ignored: a group formed on the takeoff is recognised only after the exit from $R_{\text{dep}}$ (Section 4.1);
- grid positions are interpolated across at most $h_{\max}$, so tracks logged more coarsely enter with gaps (Section 6.1);
- the signed median delay leaves out the instants of search, where the delay is not determined (Section 4.5);
- the null model assumes that surrogate pairs meet by chance as often as real pairs; with long shifts the weather changes (Section 7.1), so what it calls real includes pilots who share the same thermals at the same time (`method_assessment.pdf`);

- the errors of the phase labelling reach the heading condition: a climb labelled as glide compares headings that turn, and joining the short gaps mitigates the problem.

# A Why $m_{ij}(t) = 1$ does not give $m_{ji}(t) = 1$

$m_{ij}(t)$ of Definition 4.2 asks whether $i$, at instant $t$, is at a place that $j$ visits within $\delta$; $m_{ji}(t)$ asks whether $j$, at the same instant, is at a place that $i$ visits within $\delta$. These are two questions about two different pairs of instants. Both are read from one set of correspondences $\mathcal{M}_{ij}$, one on each axis, and they agree only if that set is symmetric about the diagonal $t' = t$. What does hold:

- if $m_{ij}(t) = 1$ through the pair $(t, t')$, then $m_{ji}(t') = 1$: the same pair, read on the other axis, $l = t' - t$ later;
- every instant of one indicator lies within $\delta$ of an instant of the other;
- the contact $a_{ij} = m_{ij} \vee m_{ji}$ is symmetric: $a_{ij} = a_{ji}$.

This appendix keeps only the horizontal condition of $\mathcal{M}_{ij}$: the vertical and heading conditions only remove pairs. $m_{ij}$ fixes the instant of $i$ and lets that of $j$ vary within $\pm\delta$; $m_{ji}$ fixes the instant of $j$ and lets that of $i$ vary. Nothing makes the two answers equal.

## A.1 A worked example

Take $\delta = 120$ s and $\varepsilon_h = 500$ m, and ignore heights and headings. Pilot $i$ circles in a thermal at the point $P$ from 0 to 600 s; the radius of the circle is negligible against $\varepsilon_h$. Pilot $j$ flies east in a straight line at 10 m/s and is in $P$ at 500 s.

| $t$ (s) | $j$ relative to $P$ | $m_{ij}$ | $m_{ji}$ | $a_{ij}$ | what happens |
|---|---|---|---|---|---|
| 300 | 2.0 km west | 0 | 0 | 0 | $j$ needs more than 120 s to come within 500 m of $P$ |
| 380 | 1.2 km west | 1 | 0 | 1 | $i$ is where $j$ will be 120 s later; $j$ is not yet where $i$ has been |
| 500 | in $P$ | 1 | 1 | 1 | both are in $P$ |
| 560 | 0.6 km east | 1 | 0 | 1 | $i$ is where $j$ was 60 s earlier; $j$ has left |

**Table 7:** The four instants of the example. Over the whole flight $m_{ij} = 1$ from 331 to 600 s, $m_{ji} = 1$ from 451 to 549 s, and $a_{ij} = 1$ from 331 to 600 s (strict inequality: exactly 500 m does not count).

## A.2 The same example in the plane $(t, t')$

Plot every correspondence $(t, t')$ with the instant at which $i$ passes across and the one at which $j$ passes up (Figure 31a). Then $m_{ij}(t) = 1$ exactly when the *vertical* line at $t$ meets $\mathcal{M}_{ij}$, and $m_{ji}(t) = 1$ exactly when the *horizontal* line at height $t$ meets it. The two indicators are the shadows of $\mathcal{M}_{ij}$ on the two axes.

At $t = 380$ the vertical line meets $\mathcal{M}_{ij}$ (at the pair $(380, 500)$ and the ones just below it), and the horizontal line does not: $\mathcal{M}_{ij}$ has no point with $t' < 451$. The set is not symmetric about the diagonal. The mirror of the pair $(380, 500)$ would be $(500, 380)$, which needs $i$ at 500 s within 500 m of $j$ at 380 s, and that is 1.2 km away.

## A.3 How far apart they can be

**Claim.** If $m_{ij}(t) = 1$ there is an instant $t'$ with $|t' - t| \leq \delta$ and $m_{ji}(t') = 1$; and conversely.

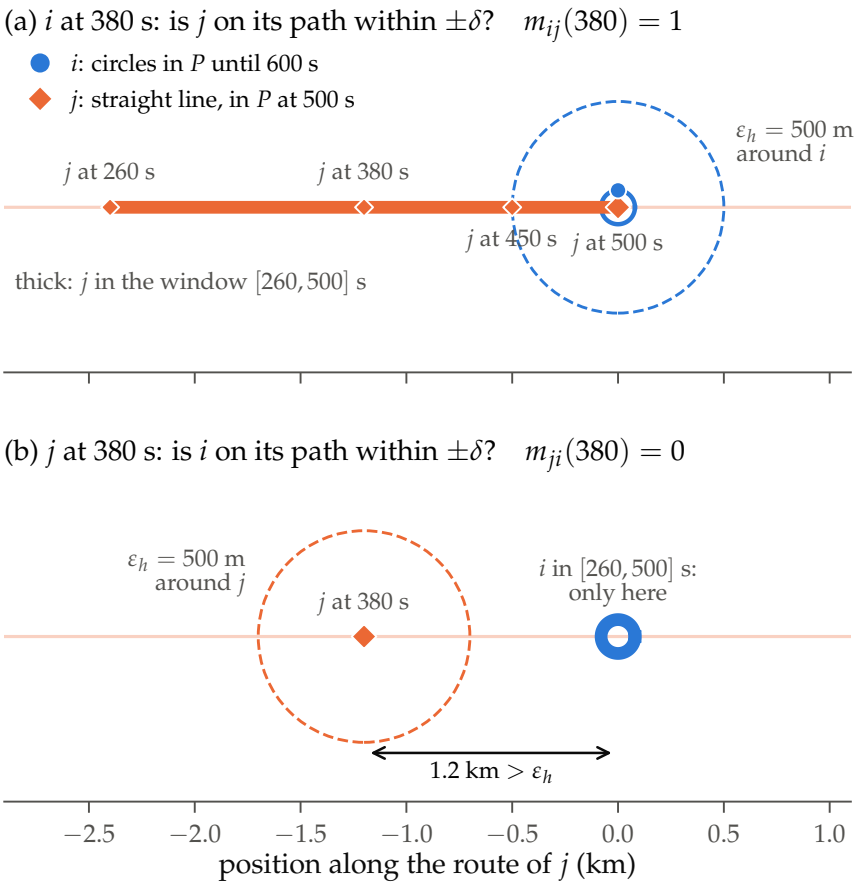



**Figure 30:** The example at 380 s. (a) $i$ is in $P$ and $j$ is 1.2 km west of it. In the window $[260, 500]$ s of $\pm\delta$ (thick), $j$ comes within $\varepsilon_h$ of $i$ at 451 s and is in $P$ at 500 s: the pair $(380, 500)$ is a correspondence, so $m_{ij}(380) = 1$. (b) At the same instant $j$ is 1.2 km from $P$, and during its window $i$ has been only in $P$: no position of $i$ lies within $\varepsilon_h$ of $j$, so $m_{ji}(380) = 0$. The ring only marks $P$.

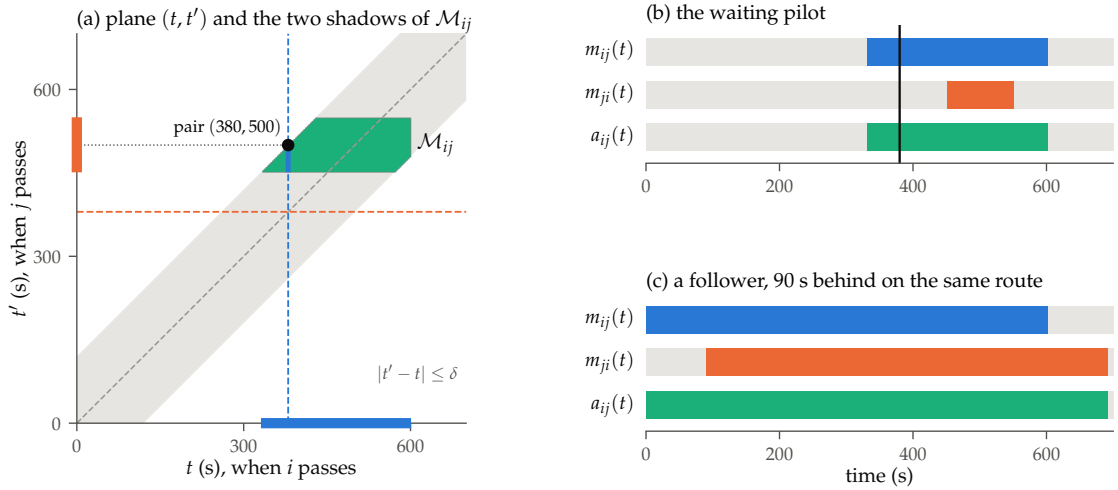



**Figure 31:** (a) The correspondences $\mathcal{M}_{ij}$ (green) in the band $|t' - t| \le \delta$, and its shadows $m_{ij}$ (blue bar on the $t$ axis) and $m_{ji}$ (orange bar on the $t'$ axis). The vertical line at 380 s meets $\mathcal{M}_{ij}$ (thick segment), the horizontal one does not. (b) The indicators over time, with the line at 380 s. (c) A pilot $j$ on the same route 90 s behind $i$: $m_{ji}$ is $m_{ij}$ delayed by exactly 90 s.

*Proof.* Take the pair $(t, t') \in \mathcal{M}_{ij}$ that gives $m_{ij}(t) = 1$. Then $|t' - t| \leq \delta$ by the definition of $\mathcal{M}_{ij}$, and $m_{ji}(t') = 1$ by the definition of $m_{ji}$. The converse is the same argument read the other way. □

So the two indicators can differ only by a shift of at most $\delta$. The bound is reached in the example: the first instant of $m_{ij}$ (331 s) is exactly $\delta$ before the first instant of $m_{ji}$ (451 s). The union $a_{ij}$ inherits it: at each end of a contact it extends the shorter indicator by up to $\delta$.

### A.4 Why the contact is the union

| definition | symmetric | in the example | remark |
|---|---|---|---|
| $m_{ij} \vee m_{ji}$ (used) | yes | 331–600 s | counts the approach and the departure within $\delta$ |
| $m_{ij} \wedge m_{ji}$ | yes | 451–549 s | shorter: each on the path of the other at once |
| $m_{ij}$ alone | no | 331–600 s, or 451–549 s for $(j, i)$ | depends on which pilot is called $i$ |

**Table 8:** Three ways to define the contact of a pair.

The union makes the contact independent of the order of the two pilots and keeps the approach and the departure, where one of them is already on the path of the other. The price is that a contact starts up to $\delta$ before the pilots are close at the same time. The mutual version $m_{ij} \wedge m_{ji}$ would give shorter contacts: the false discovery rate and $\tau^*$ would have to be computed again with the null model, which is not done here.

**In the code.** `contacts.scan` returns the instants of $m_{ij} = 1$ and, separately, those of $m_{ji} = 1$; `contacts.pair_intervals` takes their union. Scenarios: `models.WAIT`, `models.FOLLOW`; figures: `plots.asymmetry`; numbers checked in `test_note.py`.

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